

Computer Vision – Lecture 7

Sliding-Window based Object Detection

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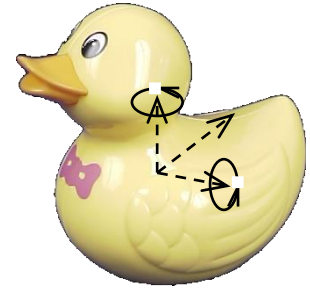
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Course Outline

- Image Processing Basics
- Segmentation
 - Segmentation and Grouping
 - Segmentation as Energy Minimization
- Recognition & Categorization
 - Sliding-Window Object Detection
- Deep Learning for Computer Vision
- 3D Reconstruction

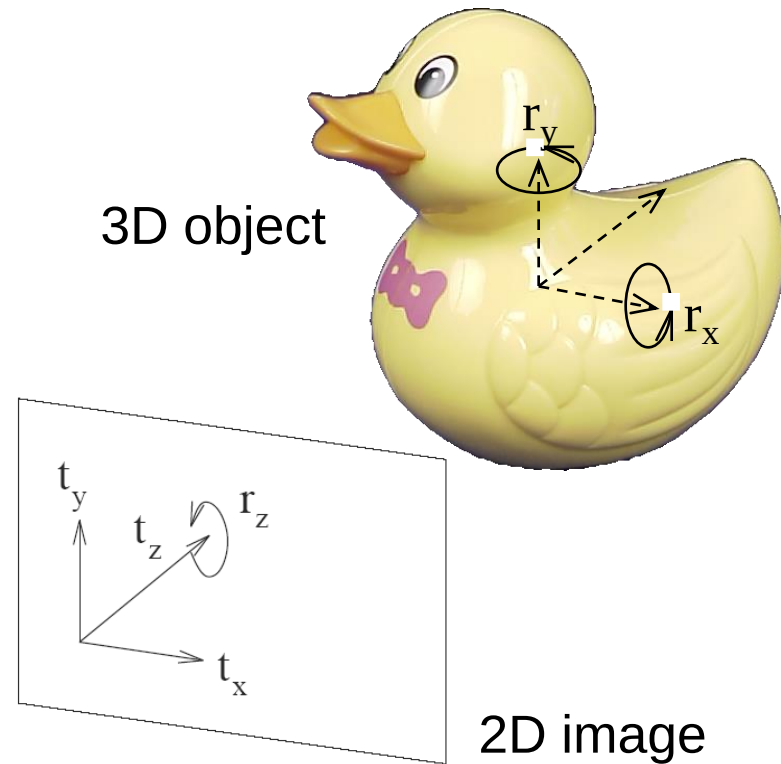
Topics of This Lecture

- Object Recognition and Categorization
 - Problem Definitions
 - Challenges
- Sliding-Window based Object Detection
 - Detection via Classification
 - Global Representations
 - Classifier Construction
- Classification with SVMs
 - Support Vector Machines
 - HOG Detector
 - DPM Detector
- Improved Search Strategies
 - Cascading
 - Region Proposal Generation



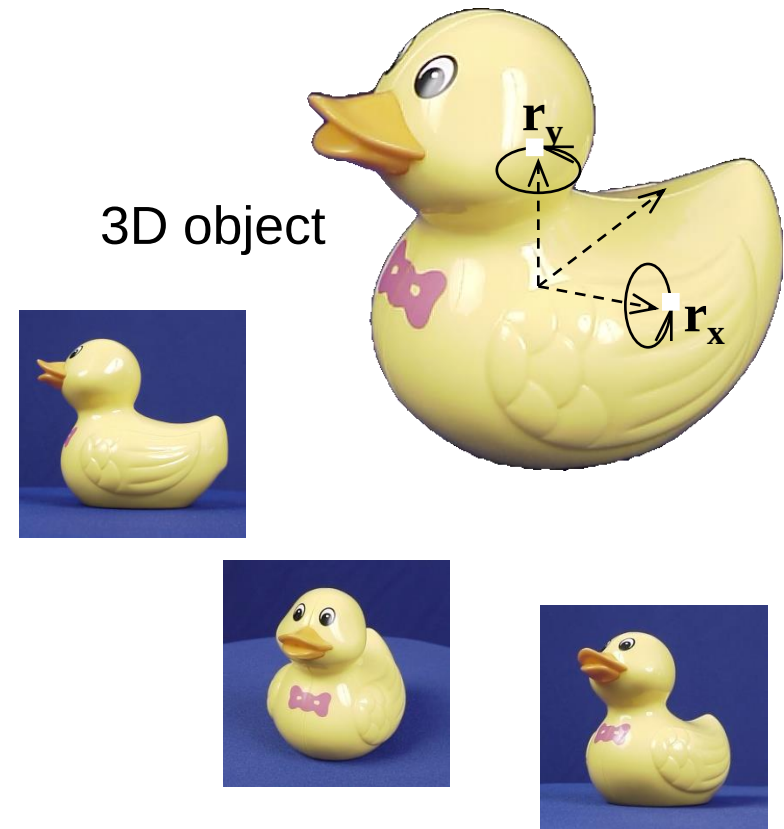
Object Recognition: Challenges

- Viewpoint changes
 - Translation
 - Image-plane rotation
 - Scale changes
 - Out-of-plane rotation
- Illumination
- Noise
- Clutter
- Occlusion



Appearance-Based Recognition

- Basic assumption
 - Objects can be represented by a set of images (“appearances”).
 - For recognition, it is sufficient to just compare the 2D appearances.
 - No 3D model is needed.

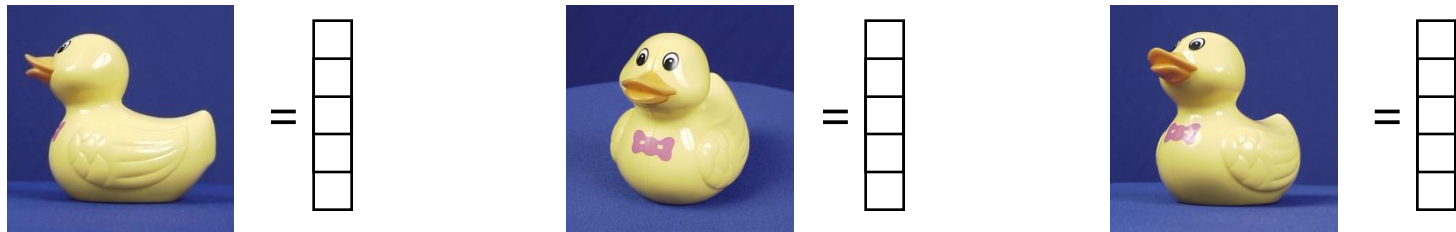


⇒ Fundamental paradigm shift in the 90's

Global Representation

- Idea

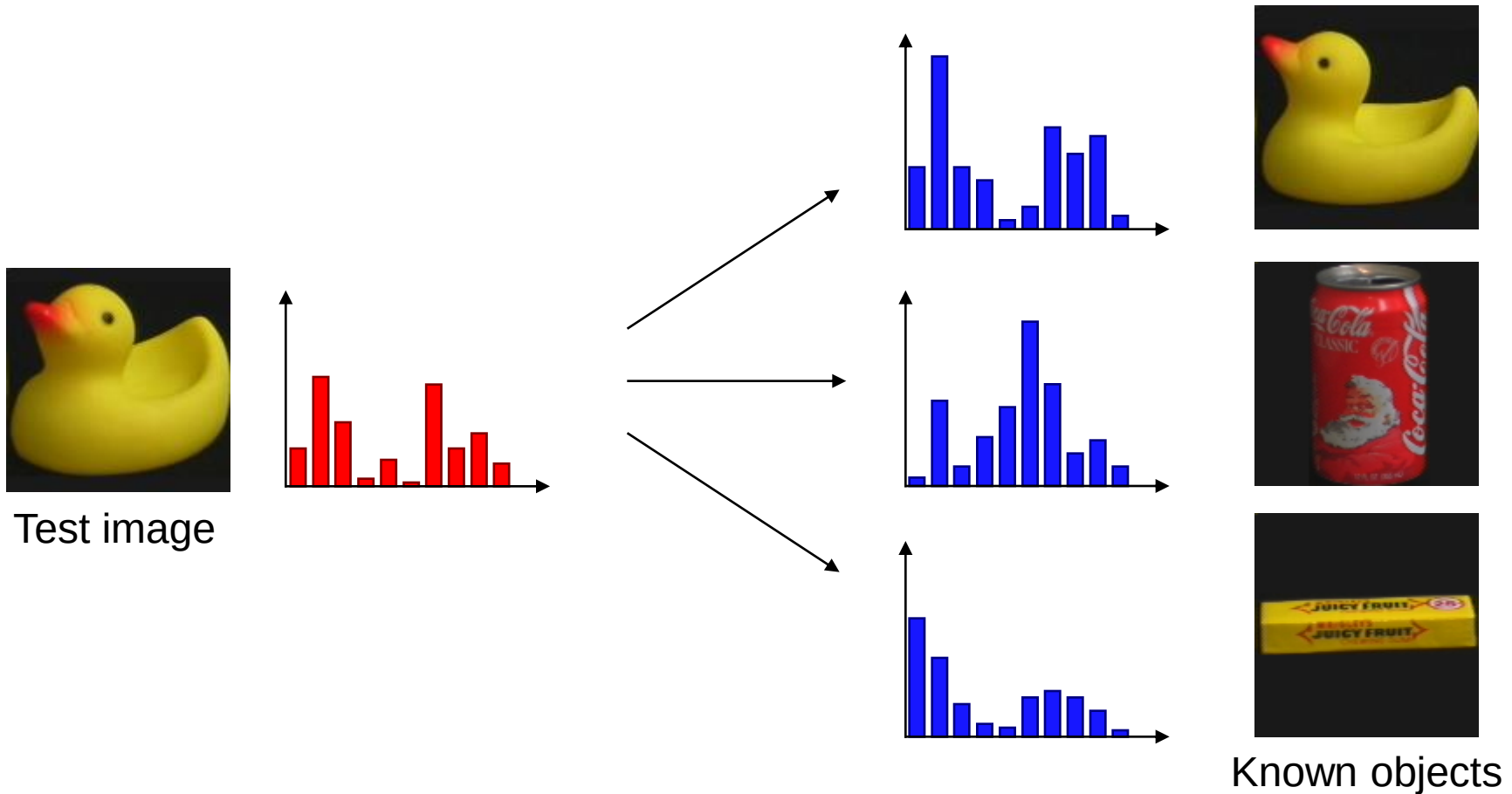
- Represent each object (view) by a global descriptor (=feature vector)



- For recognizing objects, just match the descriptors.
- Some modes of variation are built into the descriptor, the others have to be incorporated in the training data.
 - E.g., a descriptor can be made invariant to image-plane rotations.
 - Other variations:
 - Viewpoint changes
 - Translation
 - Scale changes
 - Out-of-plane rotation
 - Illumination
 - Noise
 - Clutter
 - Occlusion

Appearance based Recognition

- Recognition as feature vector matching

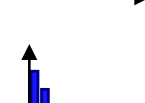
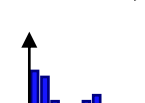
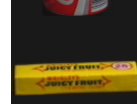
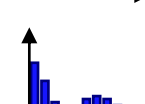
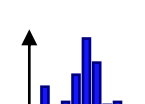
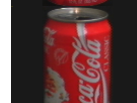
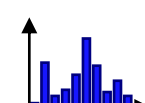
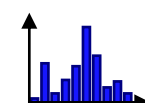
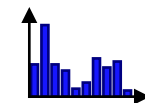
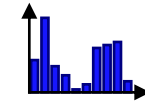
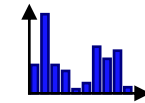
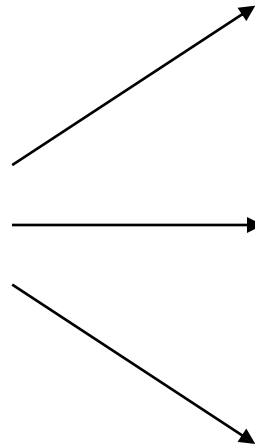
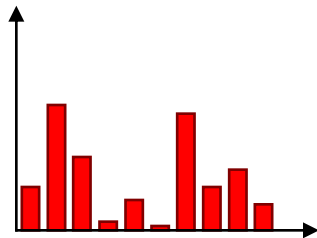


Appearance based Recognition

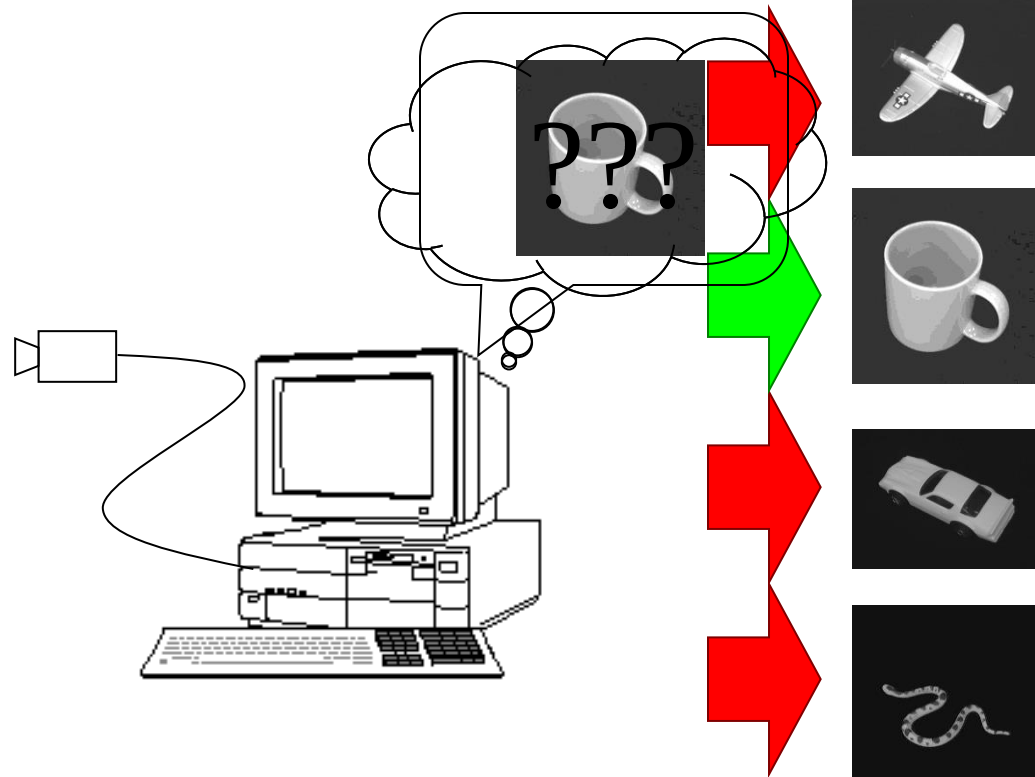
- With multiple training views



Test image



Identification vs. Categorization

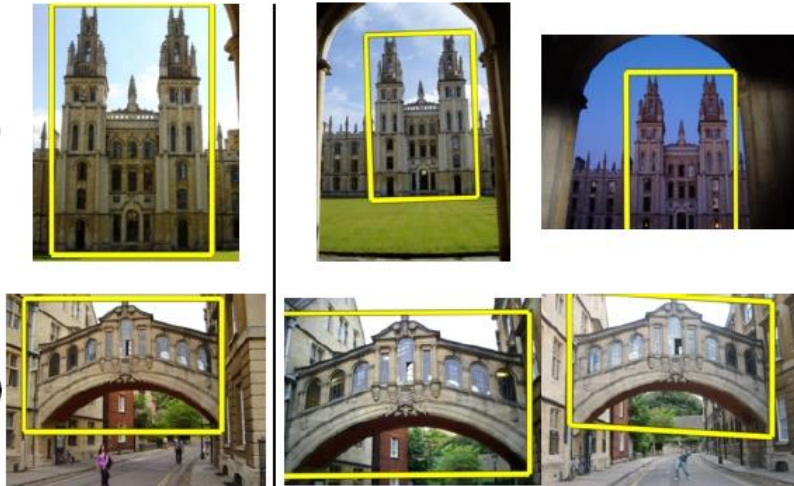


Identification vs. Categorization

- Find *this particular* object



- Recognize ANY car



- Recognize ANY cow



Object Categorization – Potential Applications

There is a wide range of applications, including...



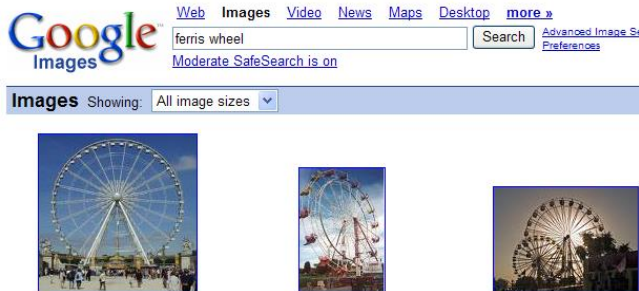
Autonomous robots



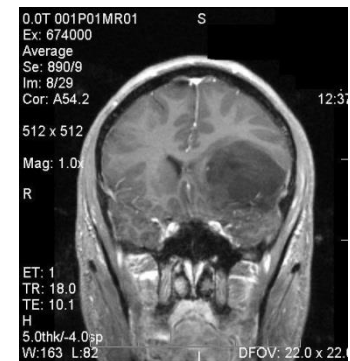
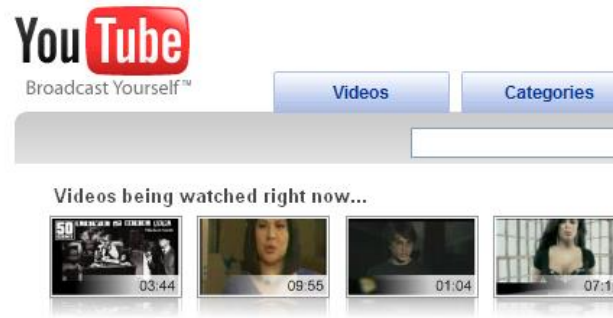
Navigation, driver safety



Consumer electronics



Content-based retrieval and analysis for
images and videos



Medical image
analysis

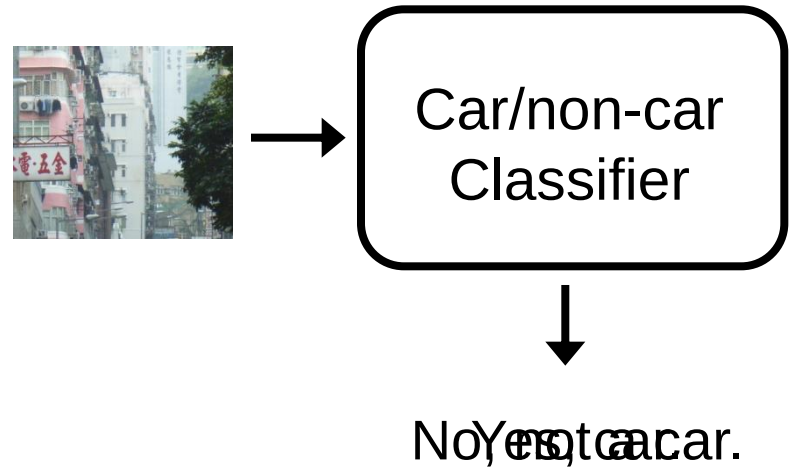
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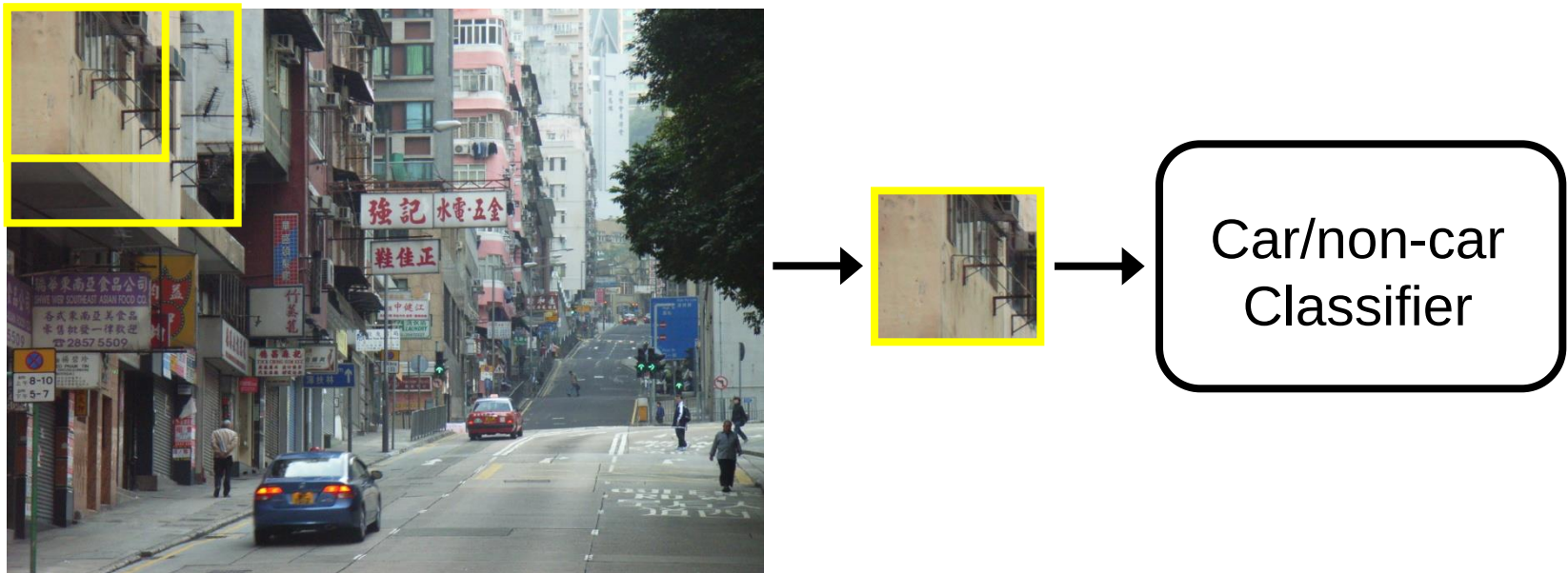
Detection via Classification: Main Idea

- Basic component: a binary classifier



Detection via Classification: Main Idea

- If the object may be in a cluttered scene, slide a window around looking for it.



- Essentially, this is a brute-force approach with many local decisions.

What is a Sliding Window Approach?

- Search over space and scale

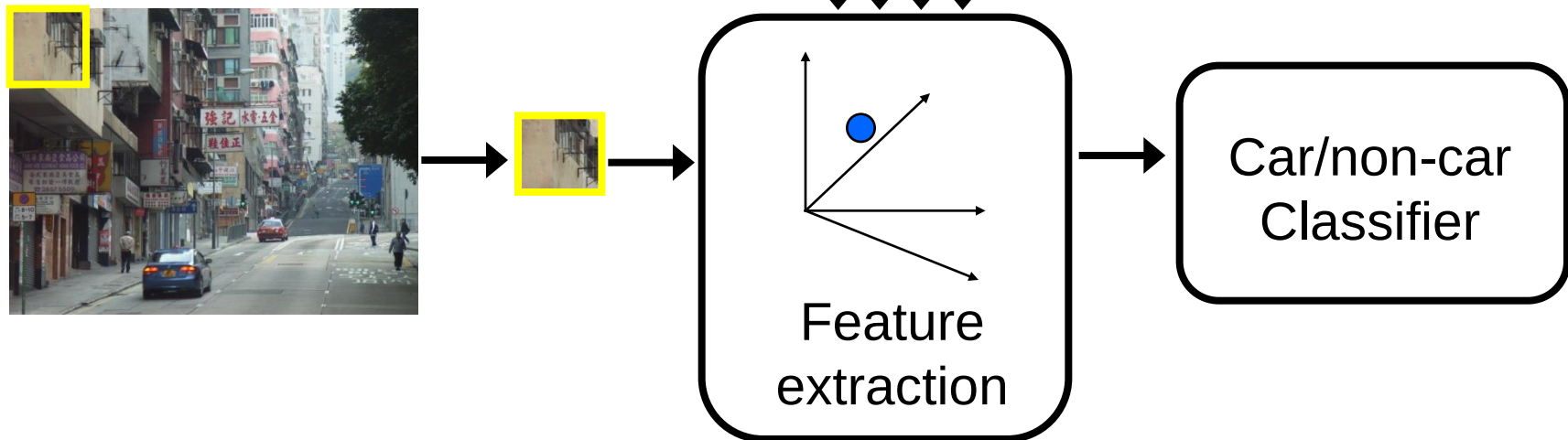
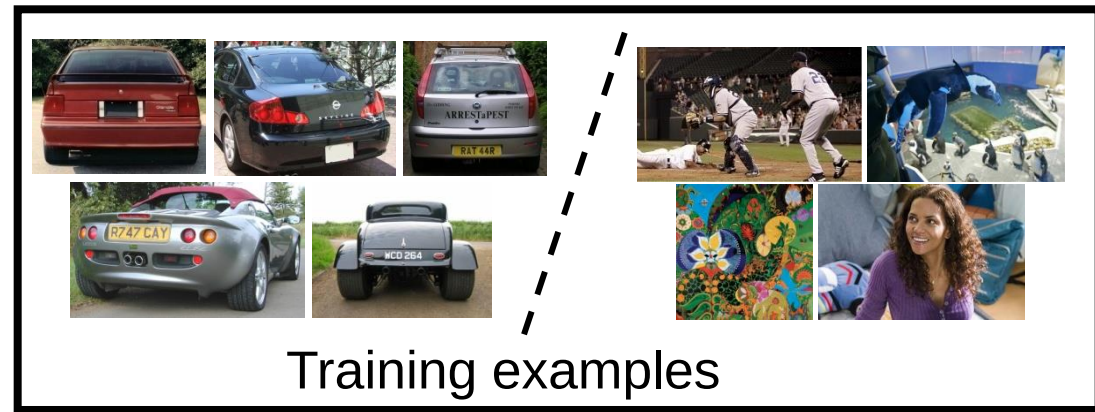


- Detection as subwindow classification problem
- *“In the absence of a more intelligent strategy, any global image classification approach can be converted into a localization approach by using a sliding-window search.”*

Detection via Classification: Main Idea

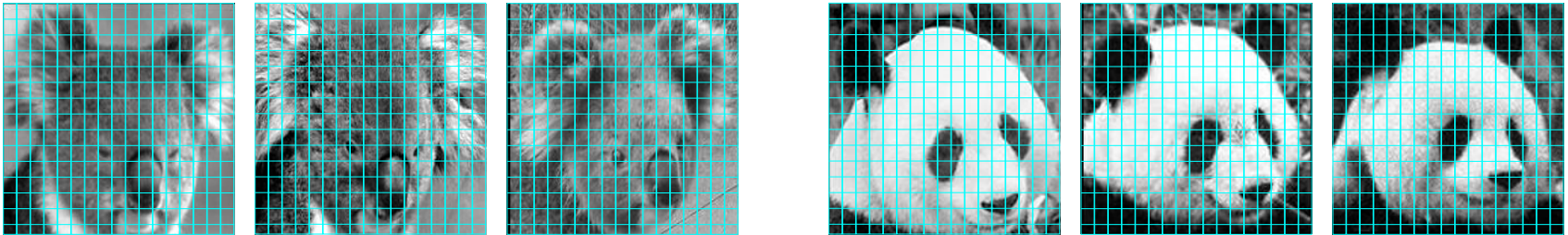
Fleshing out this pipeline a bit more, we need to:

1. Obtain training data
2. Define features
3. Define classifier



Feature Extraction: Global Appearance

- Pixel-based representations are sensitive to small shifts



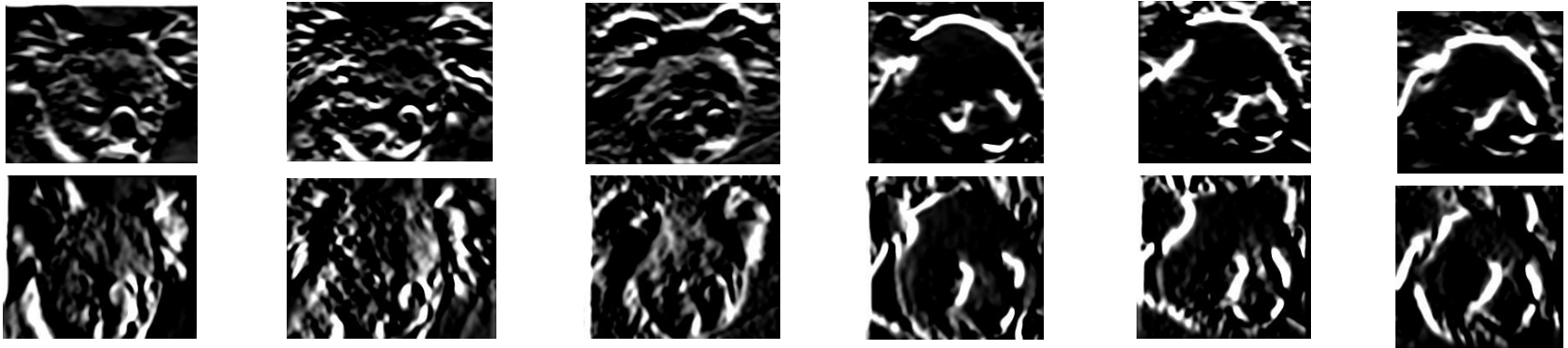
- Color or grayscale-based appearance description can be sensitive to illumination and intra-class appearance variation



Cartoon example:
an albino koala

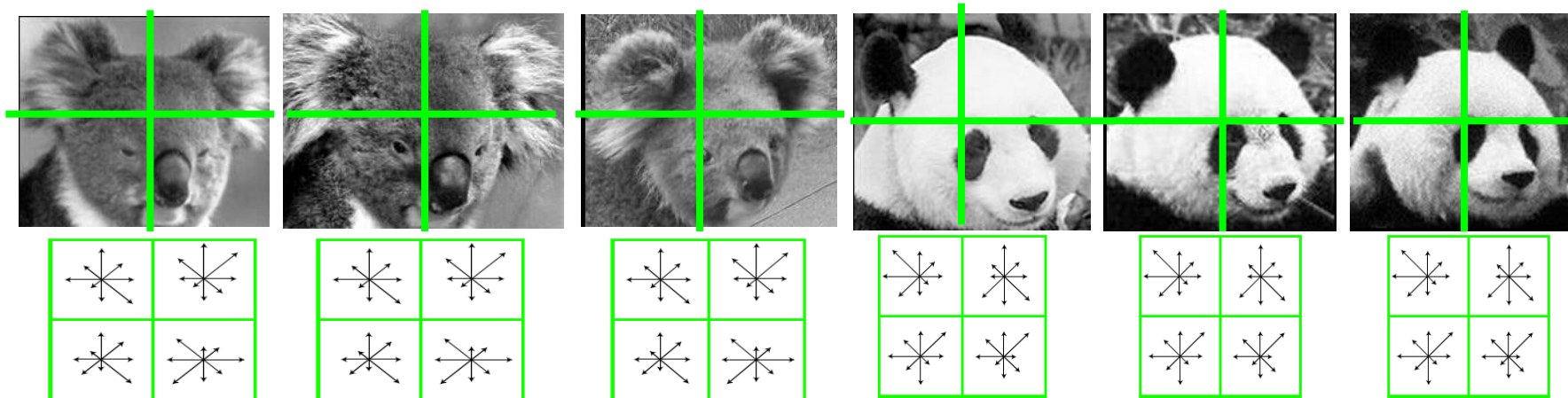
Gradient-based Representations

- Idea
 - Consider edges, contours, and (oriented) intensity gradients



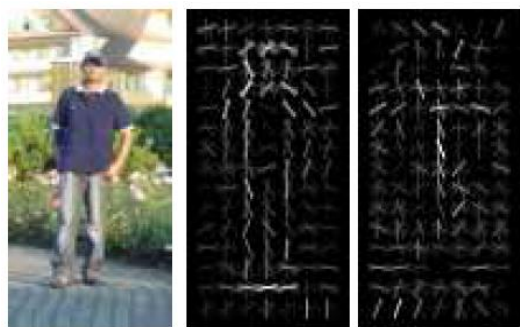
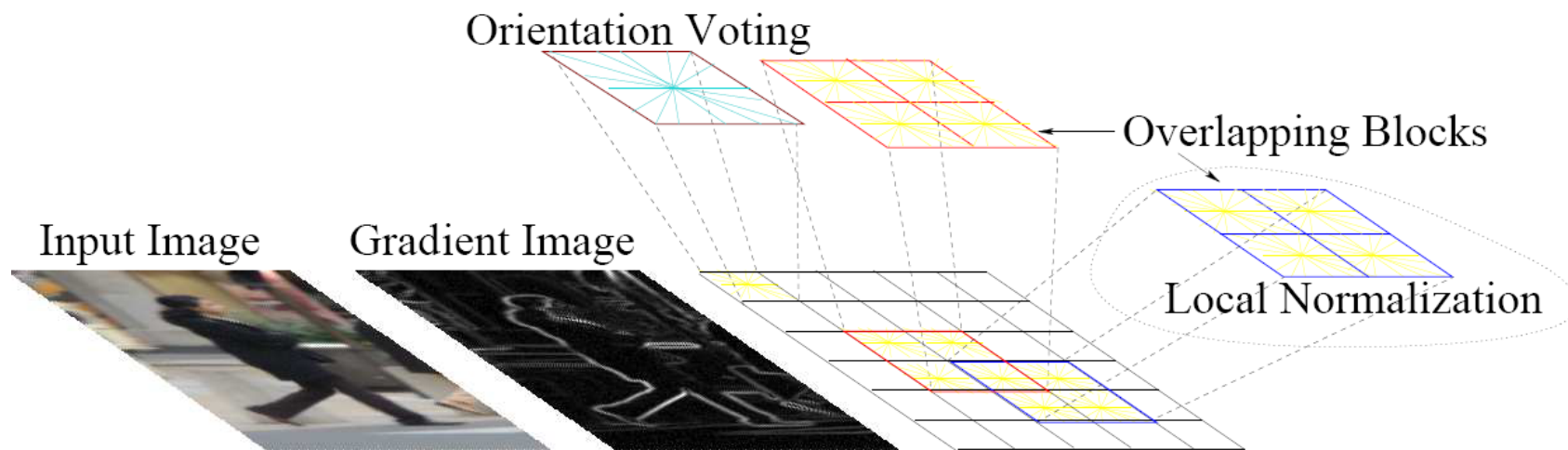
Gradient-based Representations

- Idea
 - Consider edges, contours, and (oriented) intensity gradients



- Summarize local distribution of gradients with histograms
 - Locally orderless: offers invariance to small shifts and rotations
 - Localized histograms offer more spatial information than a single global histogram (tradeoff invariant vs. discriminative)
 - Contrast-normalization: try to correct for variable illumination

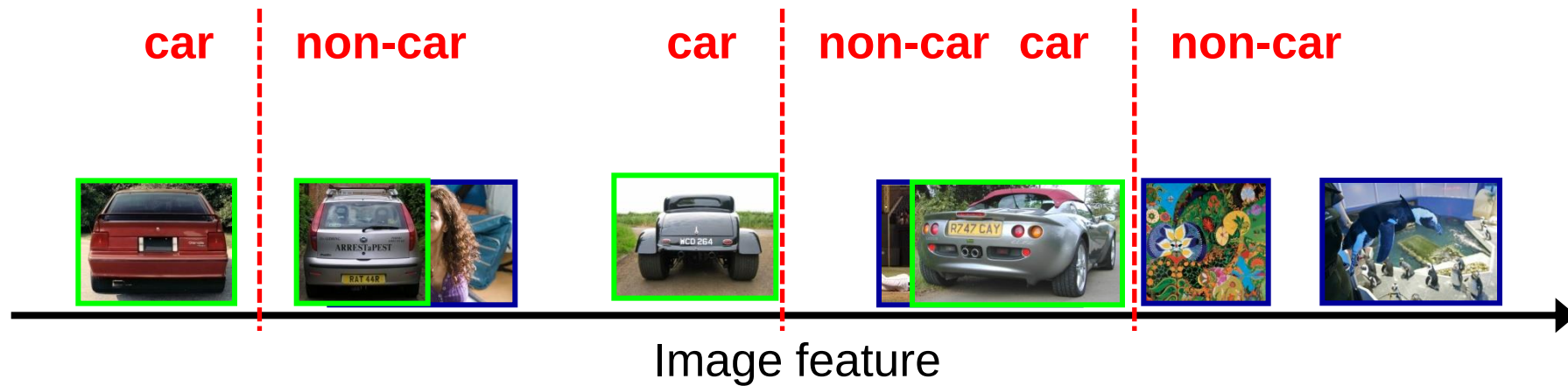
Gradient-based Representations: Histograms of Oriented Gradients (HoG)



- Map each grid cell in the input window to a histogram counting the gradients per orientation.
- Code available:
<http://pascal.inrialpes.fr/soft/olt/>

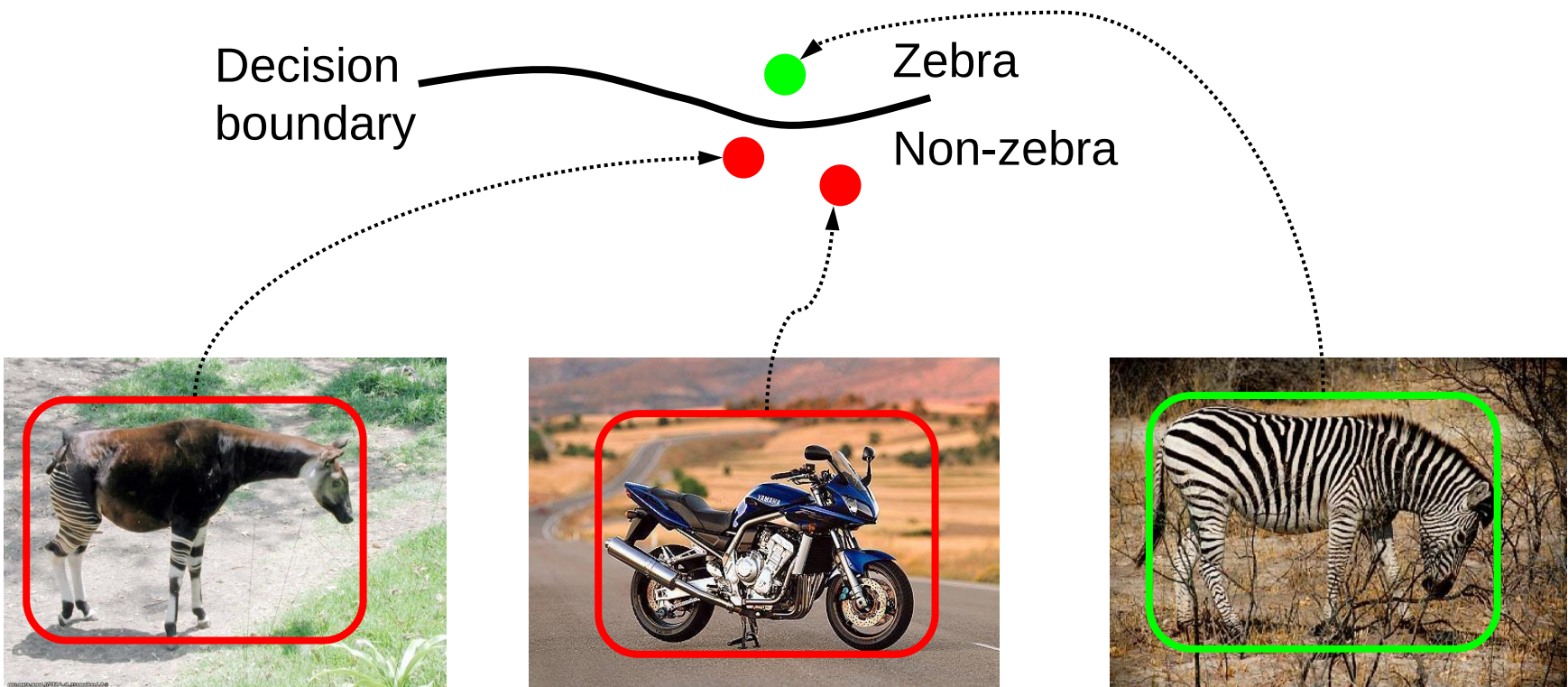
Classifier Construction

- How to compute a decision for each subwindow?



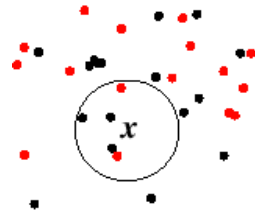
Discriminative Methods

- Learn a decision rule (classifier) assigning image features to different classes



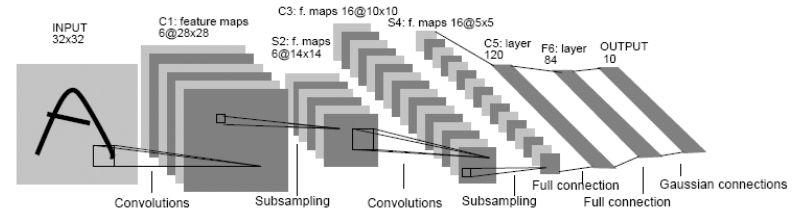
Classifier Construction: Many Choices...

Nearest Neighbor



Berg, Berg, Malik 2005,
Chum, Zisserman 2007,
Boiman, Shechtman, Irani 2008, ...

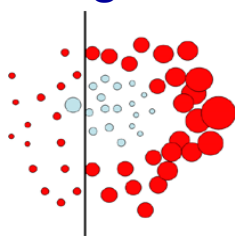
Neural networks



LeCun, Bottou, Bengio, Haffner 1998
Rowley, Baluja, Kanade 1998

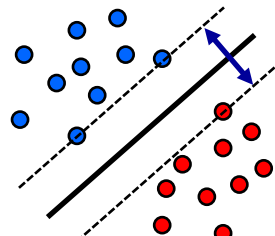
...

Boosting



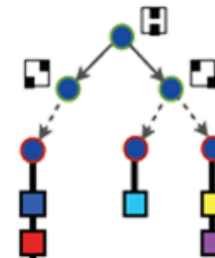
Viola, Jones 2001,
Torralba et al. 2004,
Opelt et al. 2006,
Benenson 2012, ...

Support Vector Machines



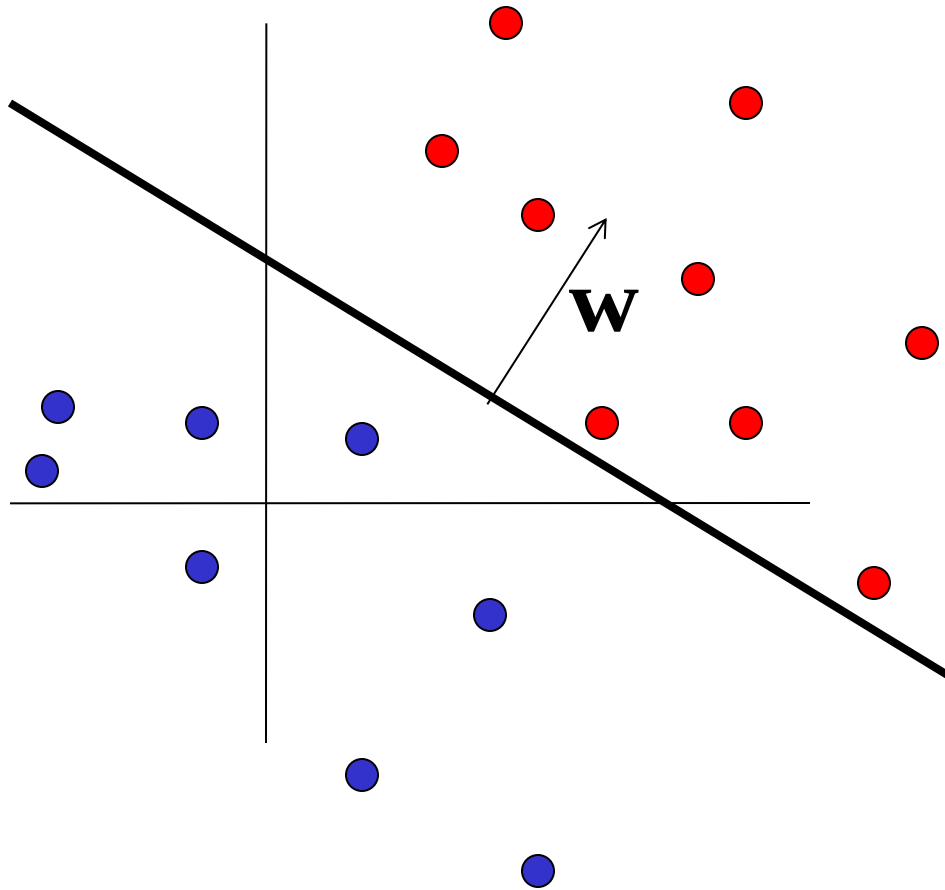
Vapnik, Schölkopf 1995,
Papageorgiou, Poggio '01,
Dalal, Triggs 2005,
Vedaldi, Zisserman 2012

Randomized Forests



Amit, Geman 1997,
Breiman 2001,
Lepetit, Fua 2006,
Gall, Lempitsky 2009,...

Linear Classifiers



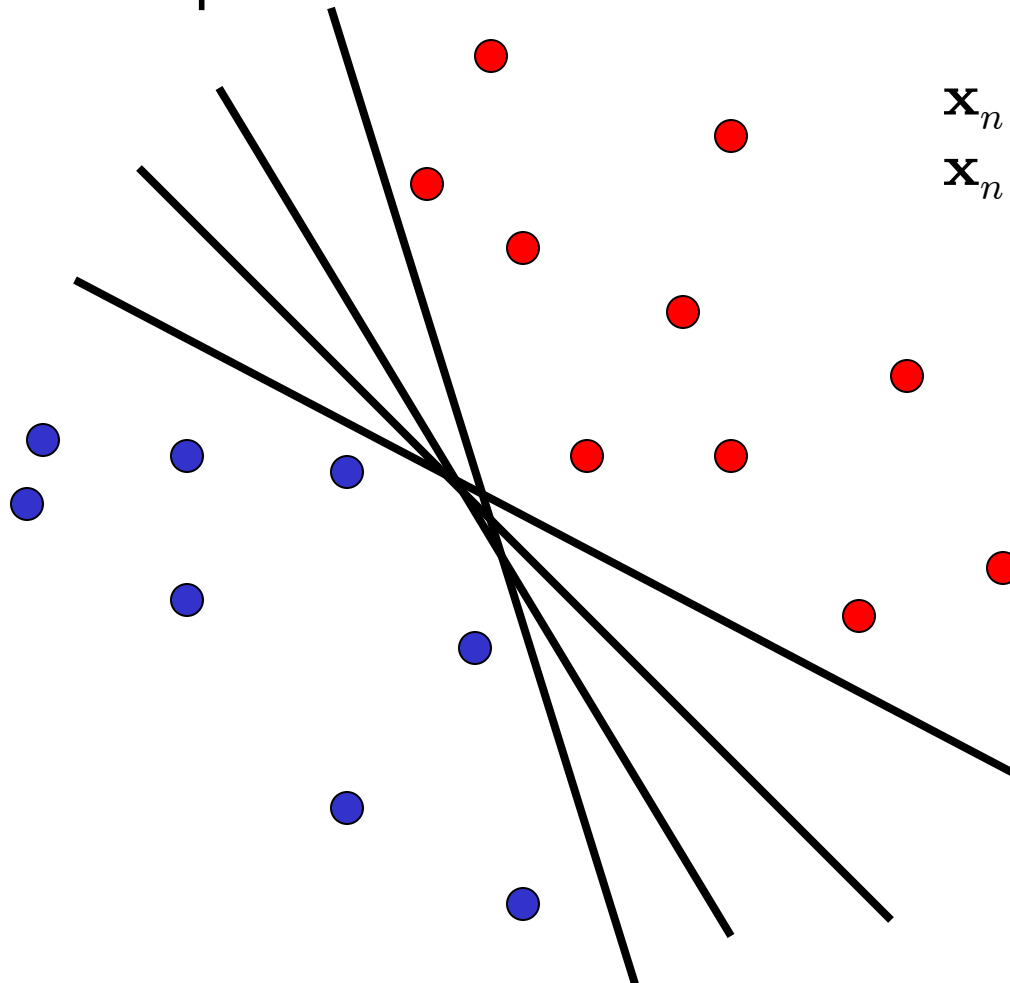
Let $\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$ $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$$w_1 x_1 + w_2 x_2 + b = 0$$

$$\mathbf{w}^T \mathbf{x} + b = 0$$

Linear Classifiers

- Find linear function to separate positive and negative examples

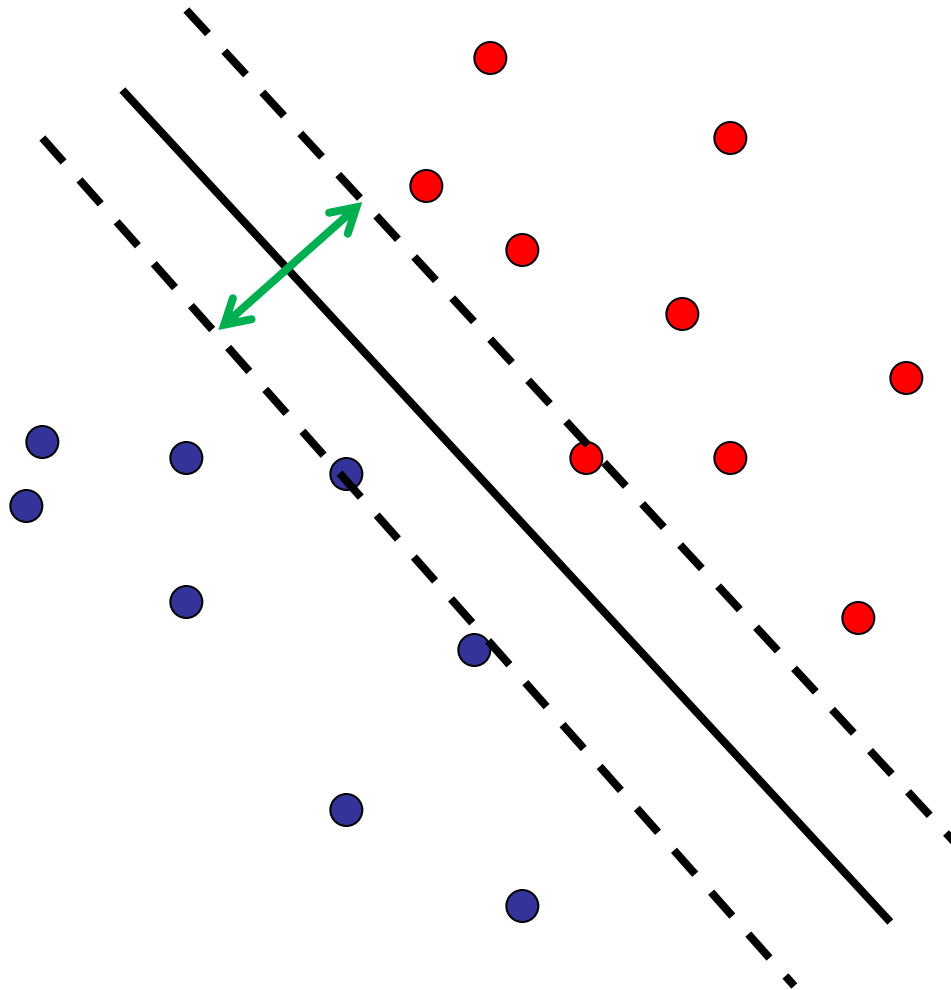


$$\mathbf{x}_n \text{ positive: } \mathbf{w}^T \mathbf{x}_n + b \geq 0$$

$$\mathbf{x}_n \text{ negative: } \mathbf{w}^T \mathbf{x}_n + b \leq 0$$

Which line
is best?

Support Vector Machines (SVMs)

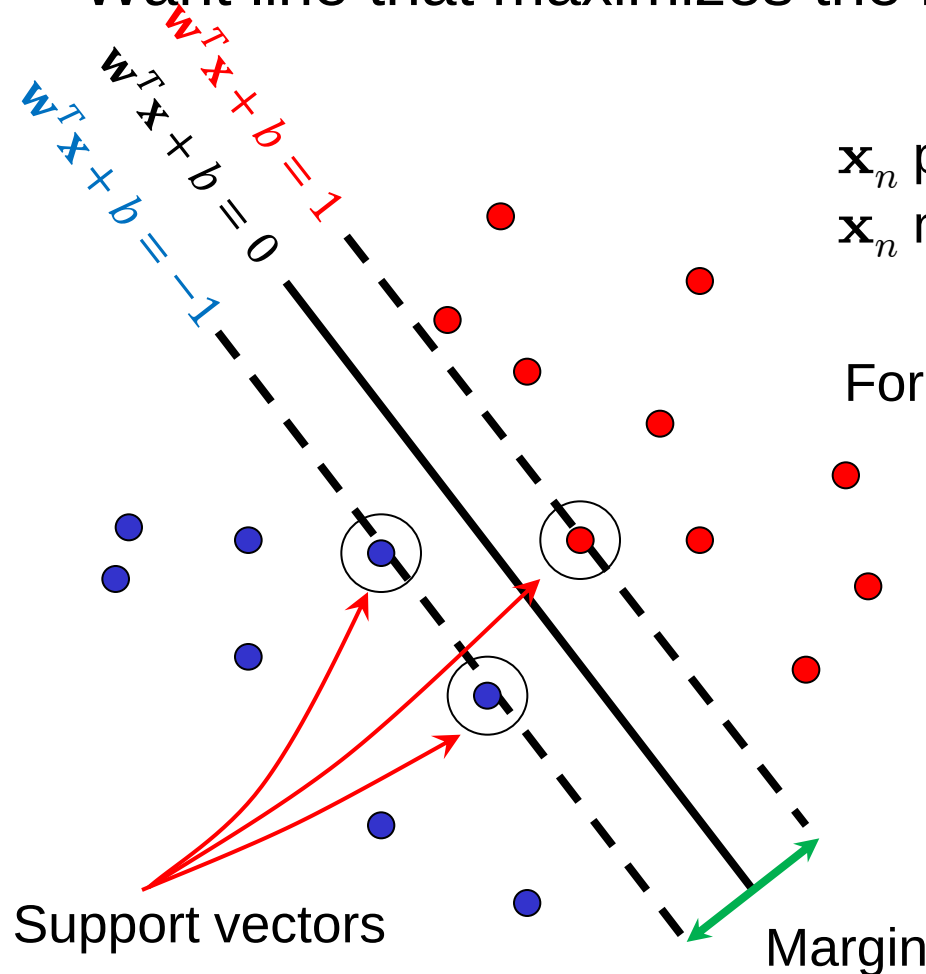


- Discriminative classifier based on *optimal separating hyperplane* (i.e. line for 2D case)
- Maximize the *margin* between the positive and negative training examples

see lecture
Machine Learning!

Support Vector Machines

- Want line that maximizes the margin.



$$\mathbf{x}_n \text{ positive } (t_n = 1): \mathbf{w}^T \mathbf{x}_n + b \geq 1$$

$$\mathbf{x}_n \text{ negative } (t_n = -1): \mathbf{w}^T \mathbf{x}_n + b \leq -1$$

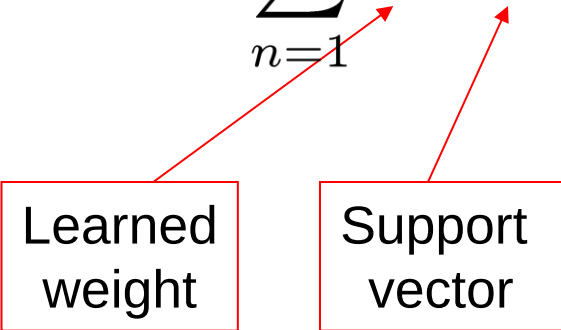
$$\text{For support, vectors, } \mathbf{w}^T \mathbf{x}_n + b = \pm 1$$

Quadratic optimization problem

$$\begin{aligned} &\text{Minimize } \frac{1}{2} \mathbf{w}^T \mathbf{w} \\ &\text{Subject to } t_n (\mathbf{w}^T \mathbf{x}_n + b) \geq 1 \end{aligned}$$

Packages available for that...

Finding the Maximum Margin Line

- Solution:
$$\mathbf{w} = \sum_{n=1}^N a_n t_n \mathbf{x}_n$$


Finding the Maximum Margin Line

- Solution:
$$\mathbf{w} = \sum_{n=1}^N a_n t_n \mathbf{x}_n$$

- Classification function:

$$f(\mathbf{x}) = \text{sign}(\mathbf{w}^T \mathbf{x} + b)$$

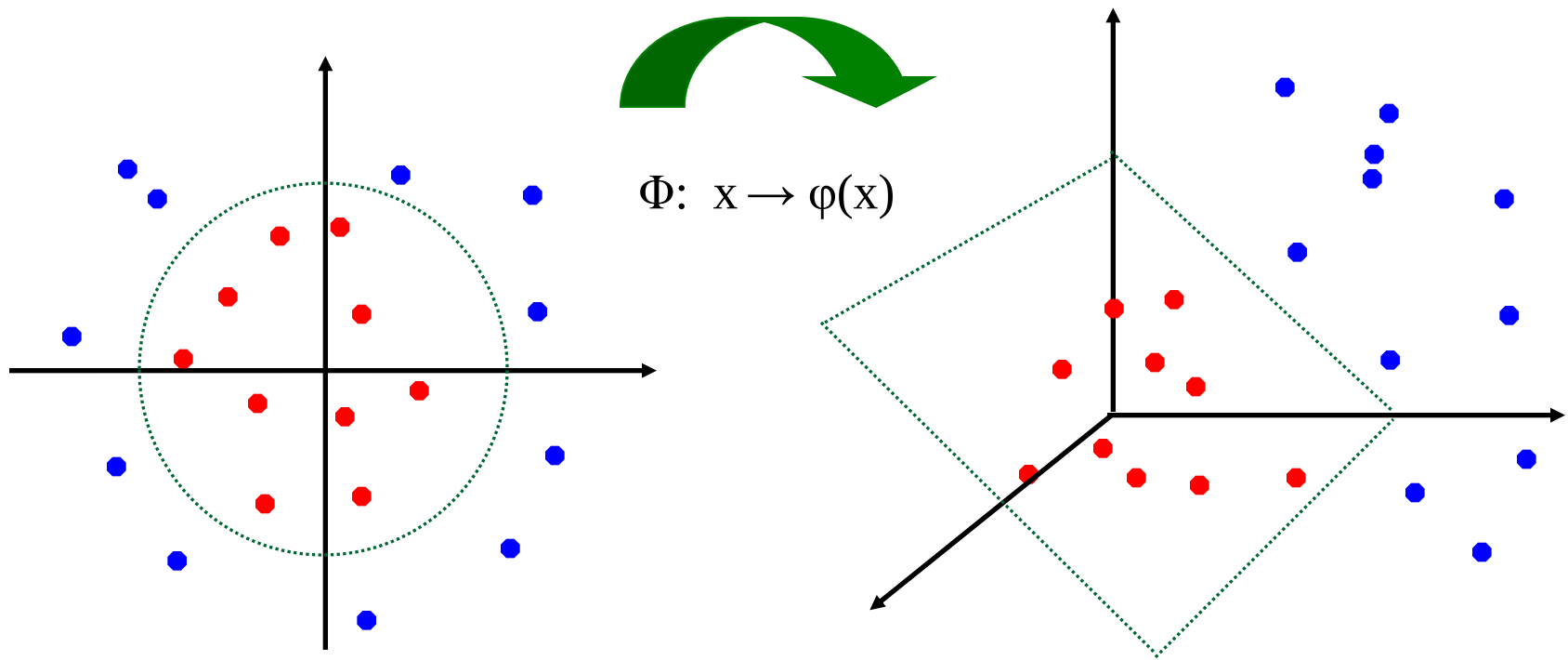
*If $f(x) < 0$, classify as neg.,
if $f(x) > 0$, classify as pos.*

$$= \text{sign} \left(\sum_{n=1}^N a_n t_n \mathbf{x}_n^T \mathbf{x} + b \right)$$

- Notice that this relies on an *inner product* between the test point $\mathbf{x}$ and the support vectors $\mathbf{x}_n$
- (Solving the optimization problem also involves computing the inner products $\mathbf{x}_n^T \mathbf{x}_m$ between all pairs of training points)

Extension: Non-Linear SVMs

- General idea: The original input space can be mapped to some higher-dimensional feature space where the training set is separable:



More on that in the Machine Learning lecture...

Nonlinear SVMs

- *The kernel trick*: instead of explicitly computing the lifting transformation $\varphi(\mathbf{x})$, define a kernel function K such that

$$K(\mathbf{x}_i, \mathbf{x}_j) = \varphi(\mathbf{x}_i) \cdot \varphi(\mathbf{x}_j)$$

- This gives a nonlinear decision boundary in the original feature space:

$$\sum_n a_n t_n K(\mathbf{x}_n, \mathbf{x}) + b$$

- Since the optimization formulation uses the data points only in the form of inner products $\varphi(\mathbf{x}_n)^T \varphi(\mathbf{x}_m)$, we never need to actually compute the lifting transformation $\varphi(\mathbf{x})$.

Some Often-Used Kernel Functions

- Linear:

$$K(\mathbf{x}_i, \mathbf{x}_j) = \mathbf{x}_i^T \mathbf{x}_j$$

- Polynomial of power p :

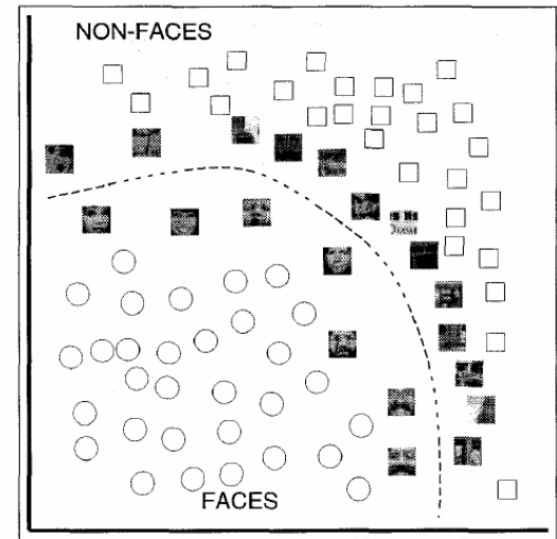
$$K(\mathbf{x}_i, \mathbf{x}_j) = (1 + \mathbf{x}_i^T \mathbf{x}_j)^p$$

- Gaussian (Radial-Basis Function):

$$K(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(-\frac{\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2\sigma^2}\right)$$

Summary: SVMs for Recognition

1. Define your representation for each example.
2. Select a kernel function.
3. Compute pairwise kernel values between labeled examples
4. Pass this “kernel matrix” to SVM optimization software to identify support vectors & weights.
5. To classify a new example: compute kernel values between new input and support vectors, apply weights, check sign of output.



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HOG Descriptor Processing Chain

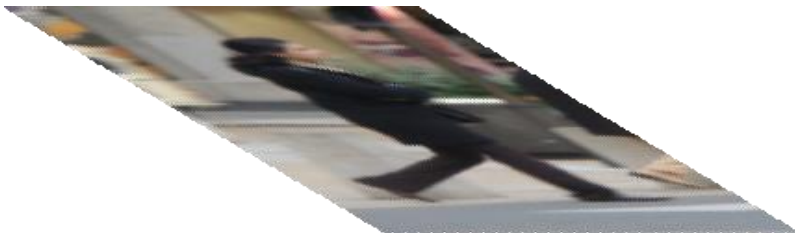


Image Window

HOG Descriptor Processing Chain

- Optional: Gamma compression
 - Goal: Reduce effect of overly strong gradients
 - Replace each pixel color/intensity by its square-root

$$x \mapsto \sqrt{x}$$

⇒ Small performance improvement



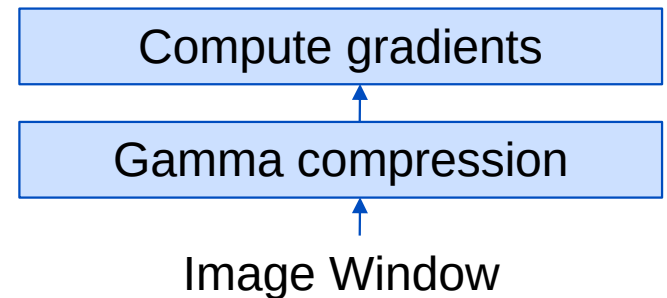
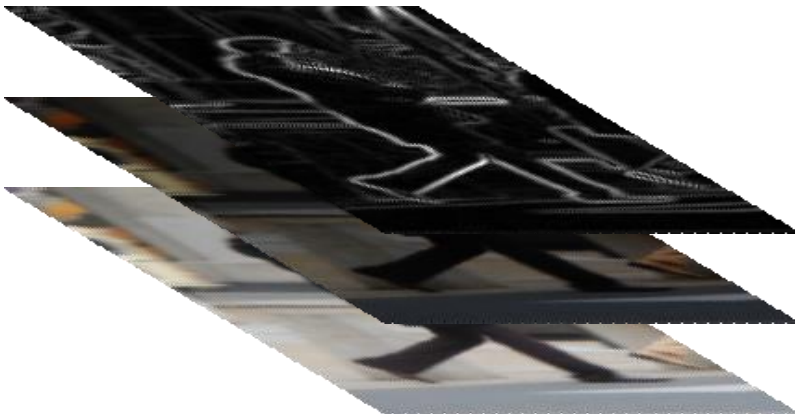
Gamma compression

Image Window

HOG Descriptor Processing Chain

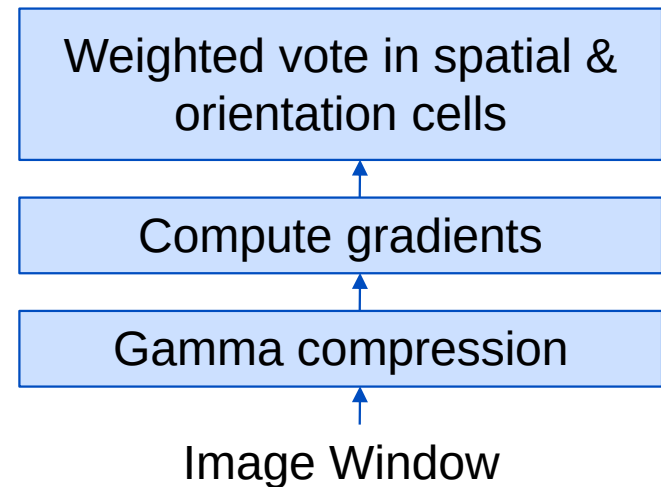
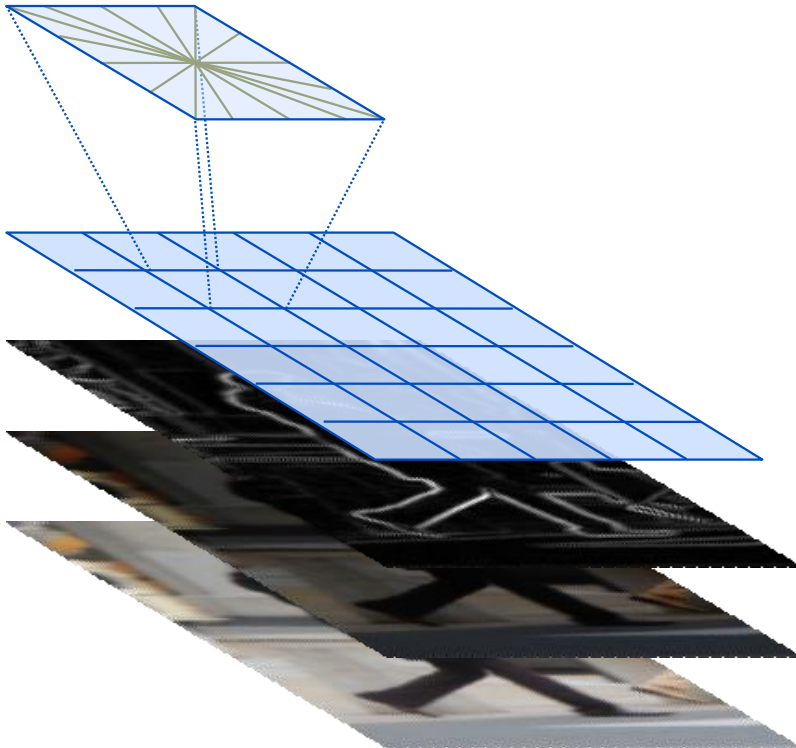
- Gradient computation
 - Compute gradients on all color channels and take strongest one
 - Simple finite difference filters work best (no Gaussian smoothing)

$$\begin{bmatrix} -1 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$



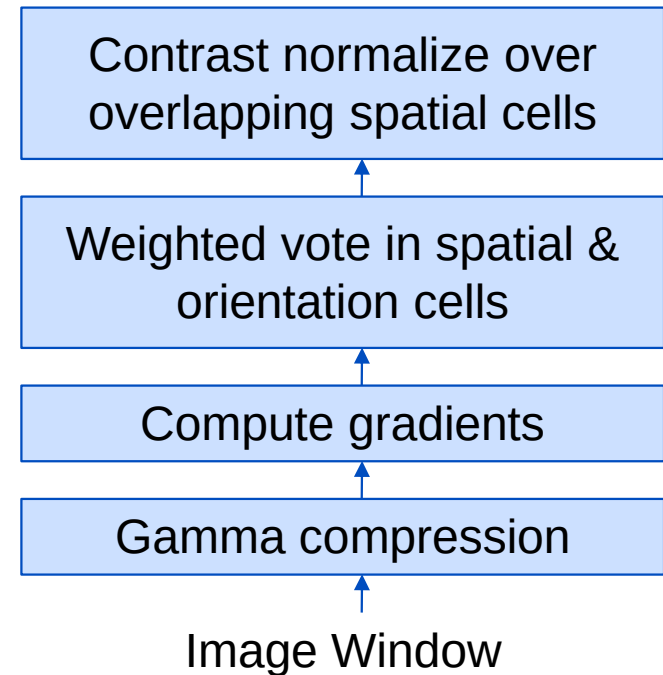
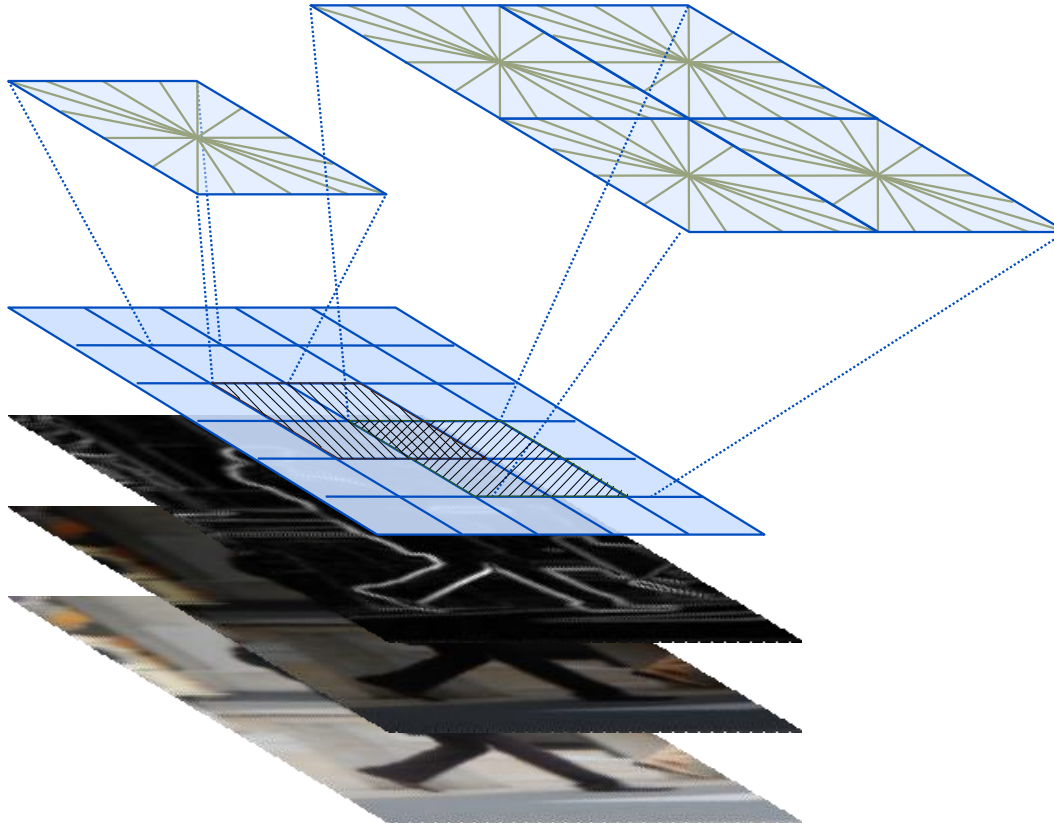
HOG Descriptor Processing Chain

- Spatial/Orientation binning
 - Compute localized histograms of oriented gradients
 - Typical subdivision:
 8×8 cells with 8 or 9 orientation bins



HOG Descriptor Processing Chain

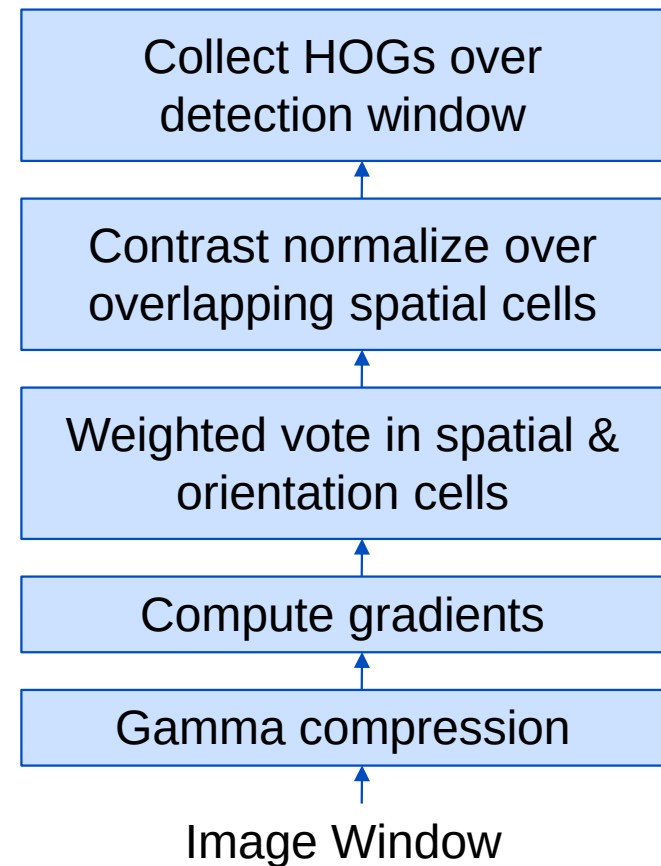
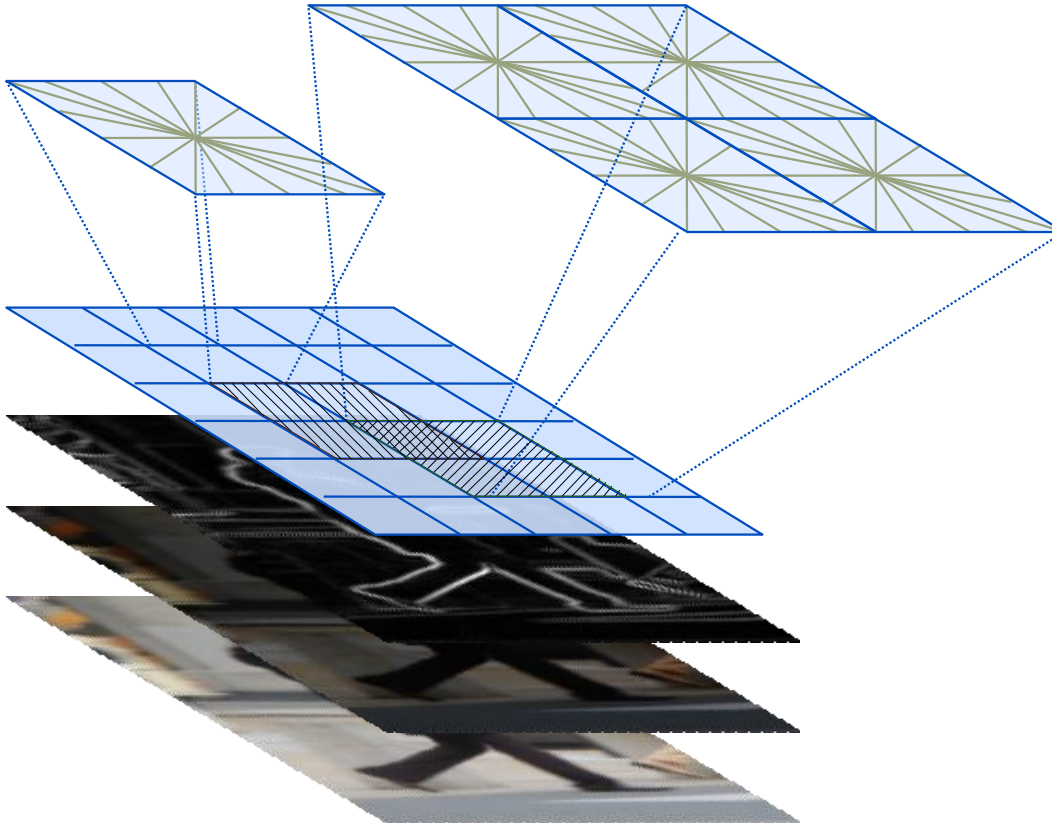
- 2-Stage contrast normalization
 - L2 normalization, clipping, L2 normalization



HOG Descriptor Processing Chain

- Feature vector construction
 - Collect HOG blocks into vector

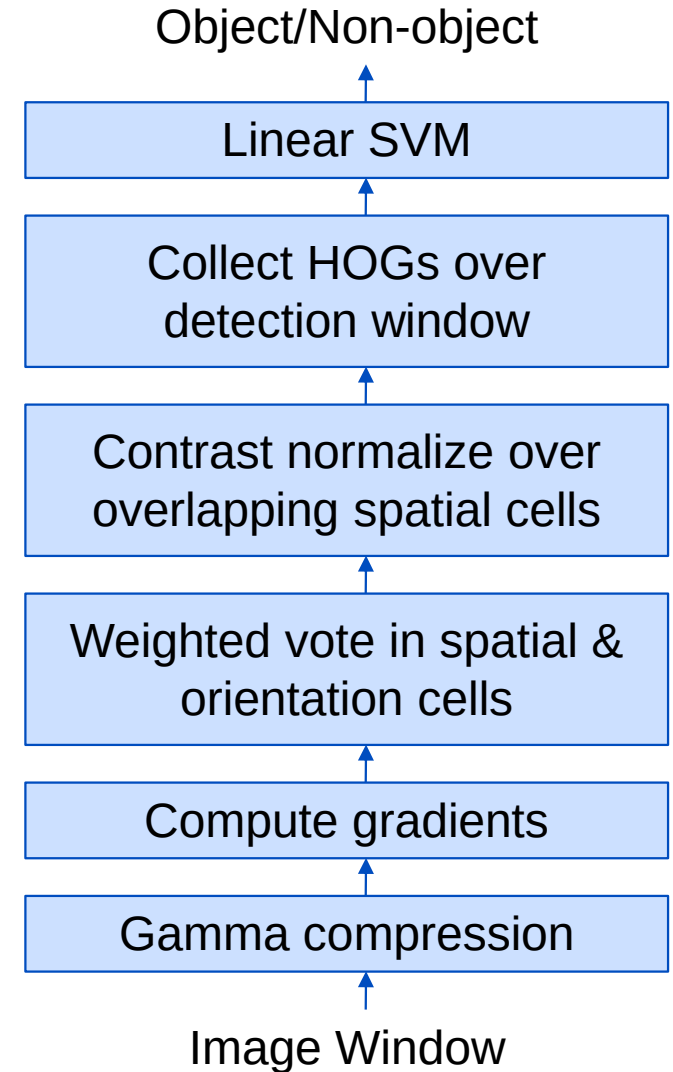
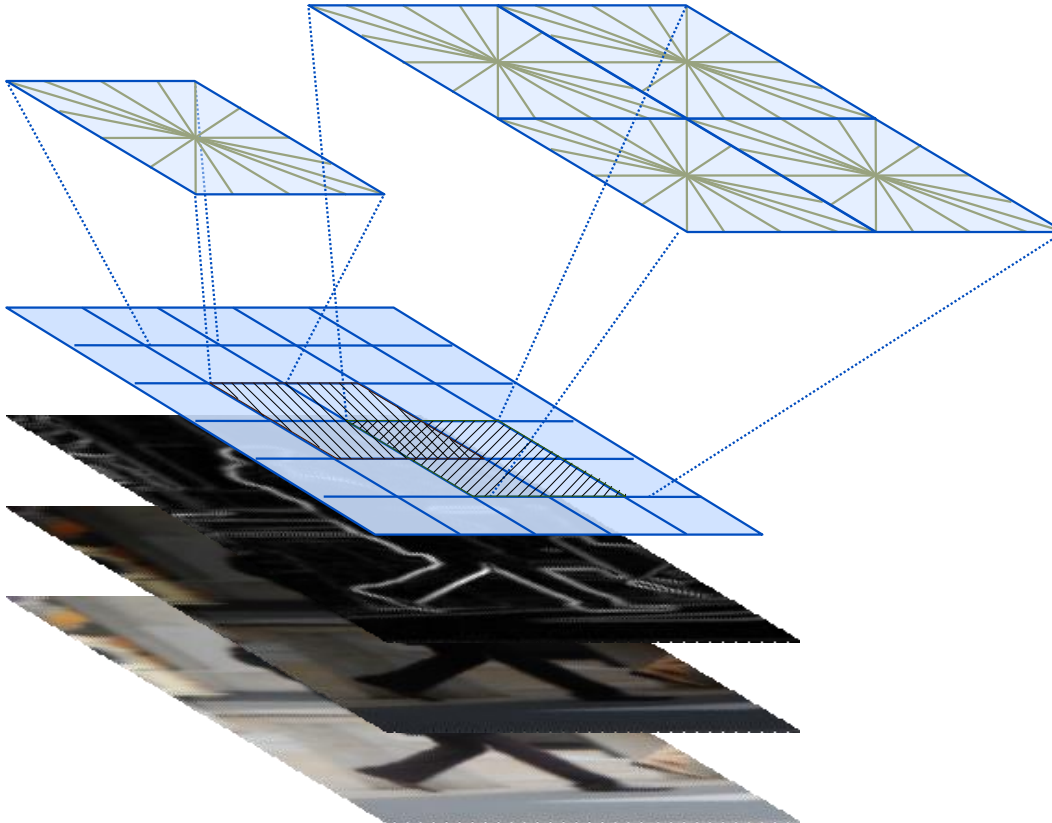
[..., ..., ..., ...]



HOG Descriptor Processing Chain

- SVM Classification
 - Typically using a linear SVM

[..., ..., ..., ...]



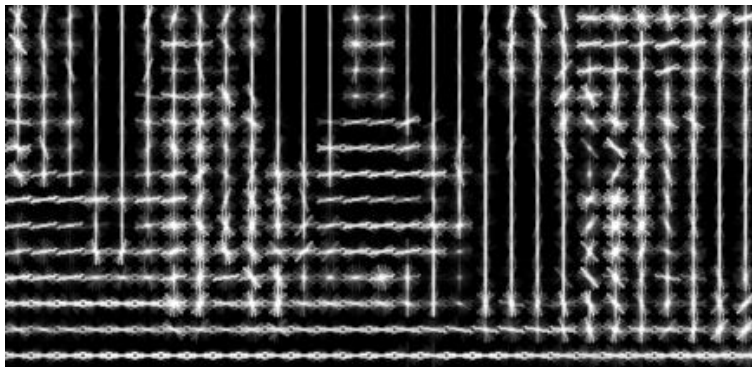
Pedestrian Detection with HOG

- Train a pedestrian template using a linear SVM
- At test time, convolve feature map with learned template $\mathbf{w}$
 - Linear SVM classification function $\Leftrightarrow$ convolution with “template” $\mathbf{w}$

$$y(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + b$$

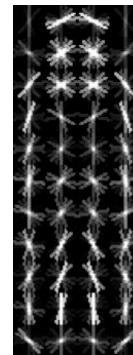
HOG feature map

$\mathbf{x}$



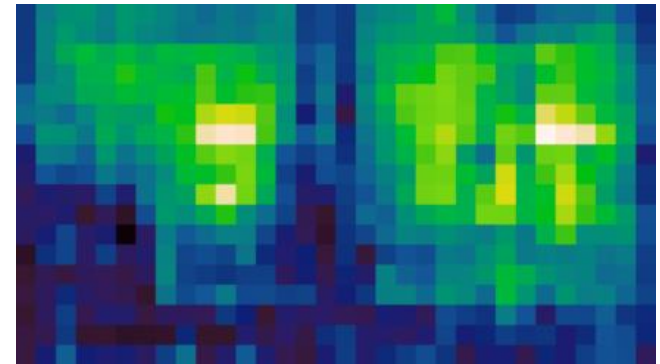
Template

$\mathbf{w}$



Detector response map

$$y(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + b$$

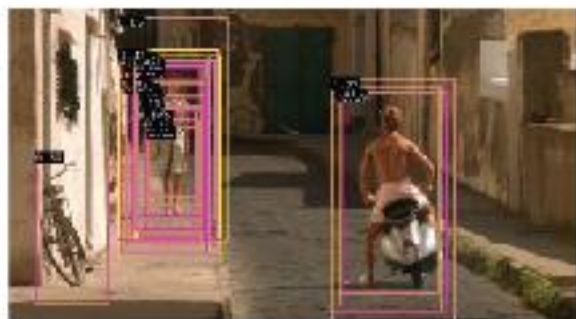


N. Dalal and B. Triggs, [Histograms of Oriented Gradients for Human Detection](#),

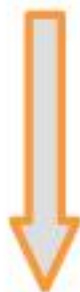
CVPR 2005

Slide credit: Svetlana Lazebnik

Non-Maximum Suppression



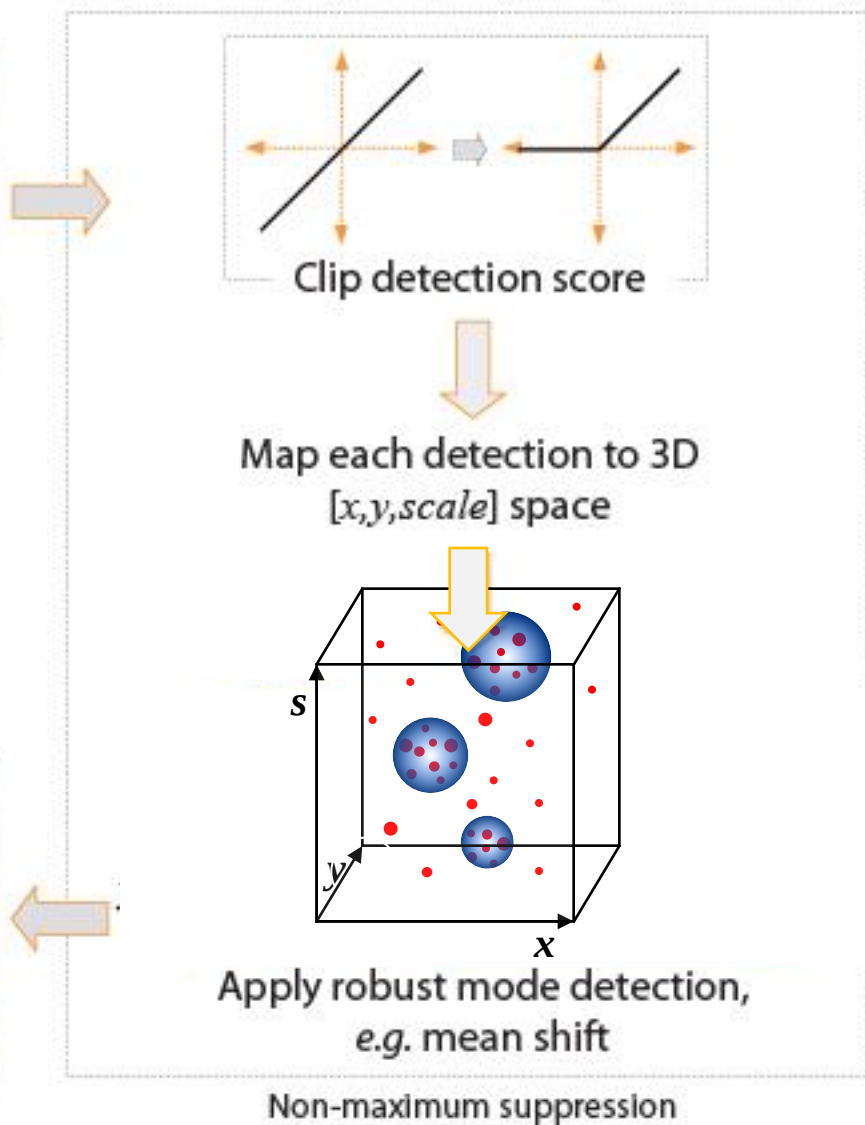
After multi-scale dense scan



Goal



Fusion of multiple detections



Pedestrian detection with HoGs & SVMs

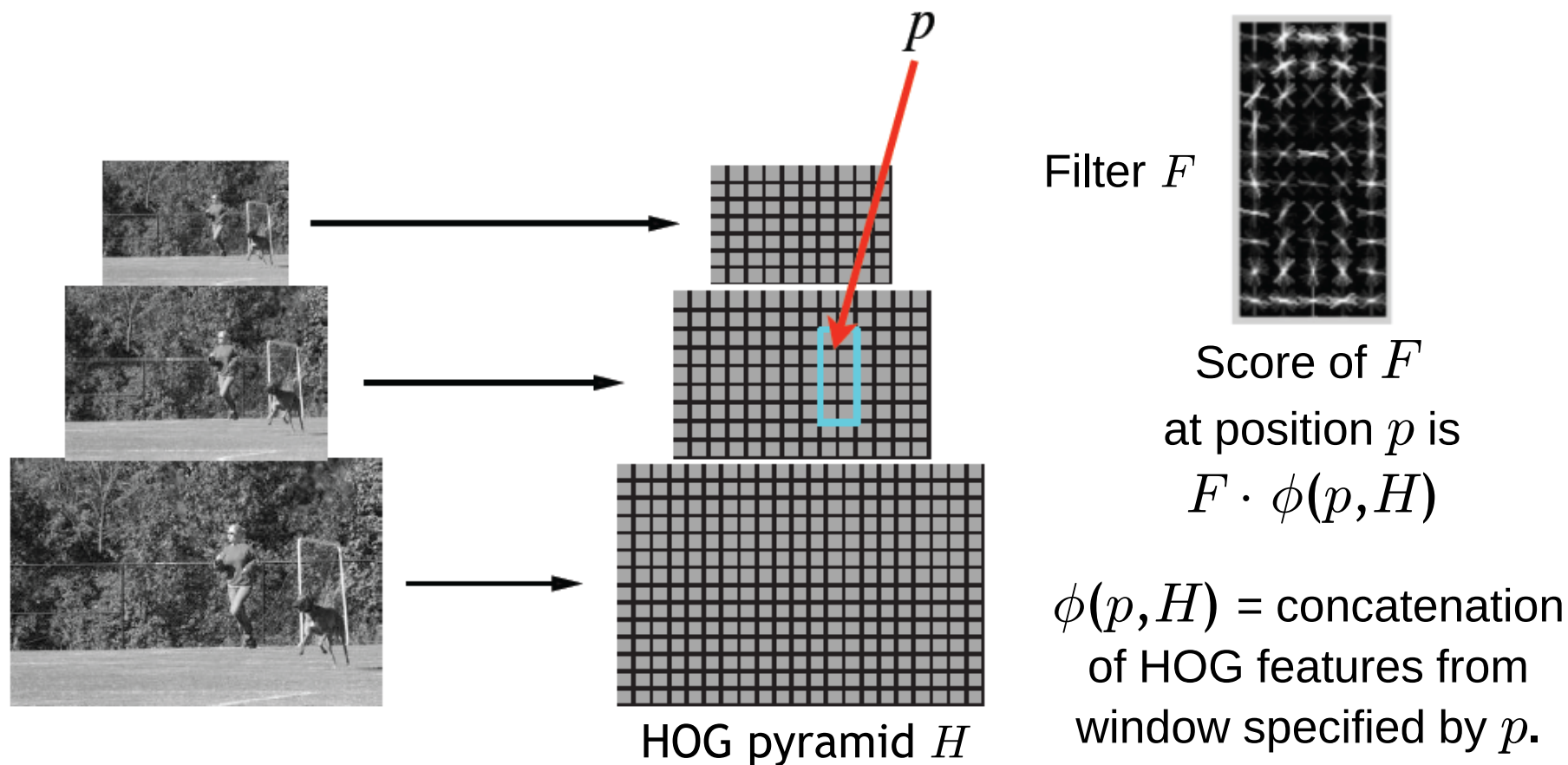


- N. Dalal, B. Triggs, [Histograms of Oriented Gradients for Human Detection](#), CVPR 2005.

Topics of This Lecture

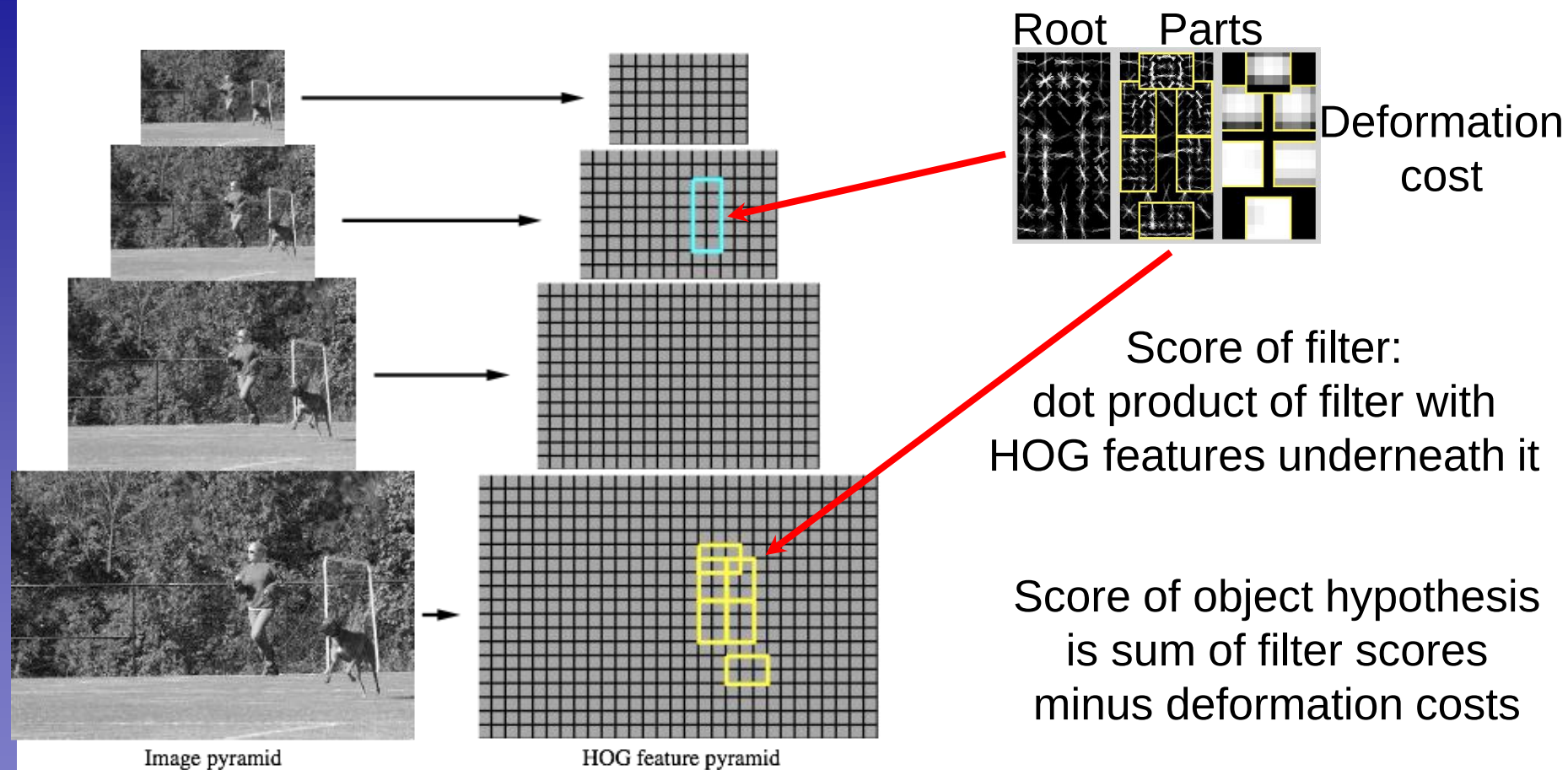
- Object Categorization
 - Problem Definition
 - Challenges
- Sliding-Window based Object Detection
 - Detection via Classification
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 - DPM Detector
- Improved Search Strategies
 - Cascading
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HOG Sliding-Window Detector



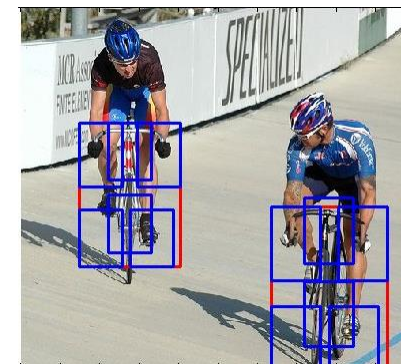
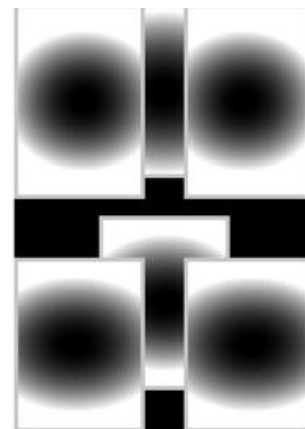
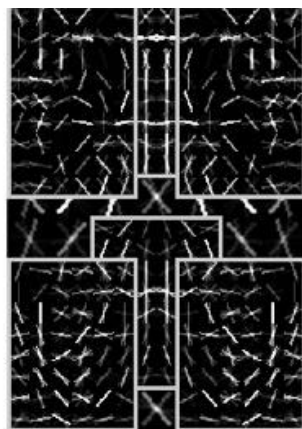
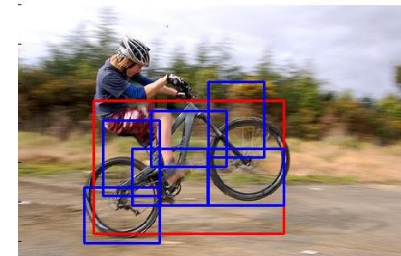
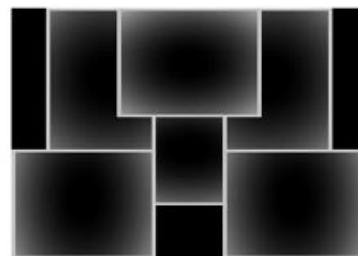
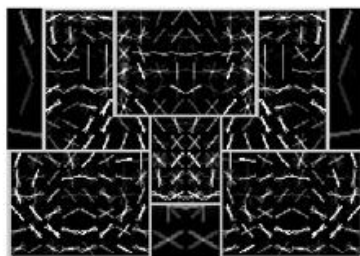
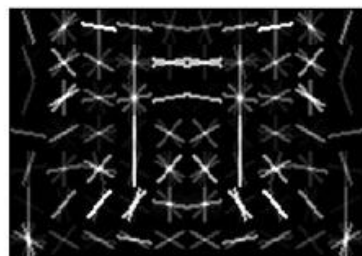
- Array of weights for features in window of HOG pyramid
- Score is dot product of filter and vector

Deformable Part-based Model (DPM)



- Multiscale model captures features at two resolutions

Example: 2-Component Bicycle Model



Root filters
coarse resolution

Part filters
finer resolution

Deformation
models

Example
detections

Score of a Hypothesis

$$\text{score}(p_0, \dots, p_n) = \sum_{i=0}^n F_i \cdot \phi(H, p_i) - \sum_{i=1}^n d_i \cdot (dx_i^2, dy_i^2)$$

“data term”

↑ filters

“spatial prior”

↑ displacements

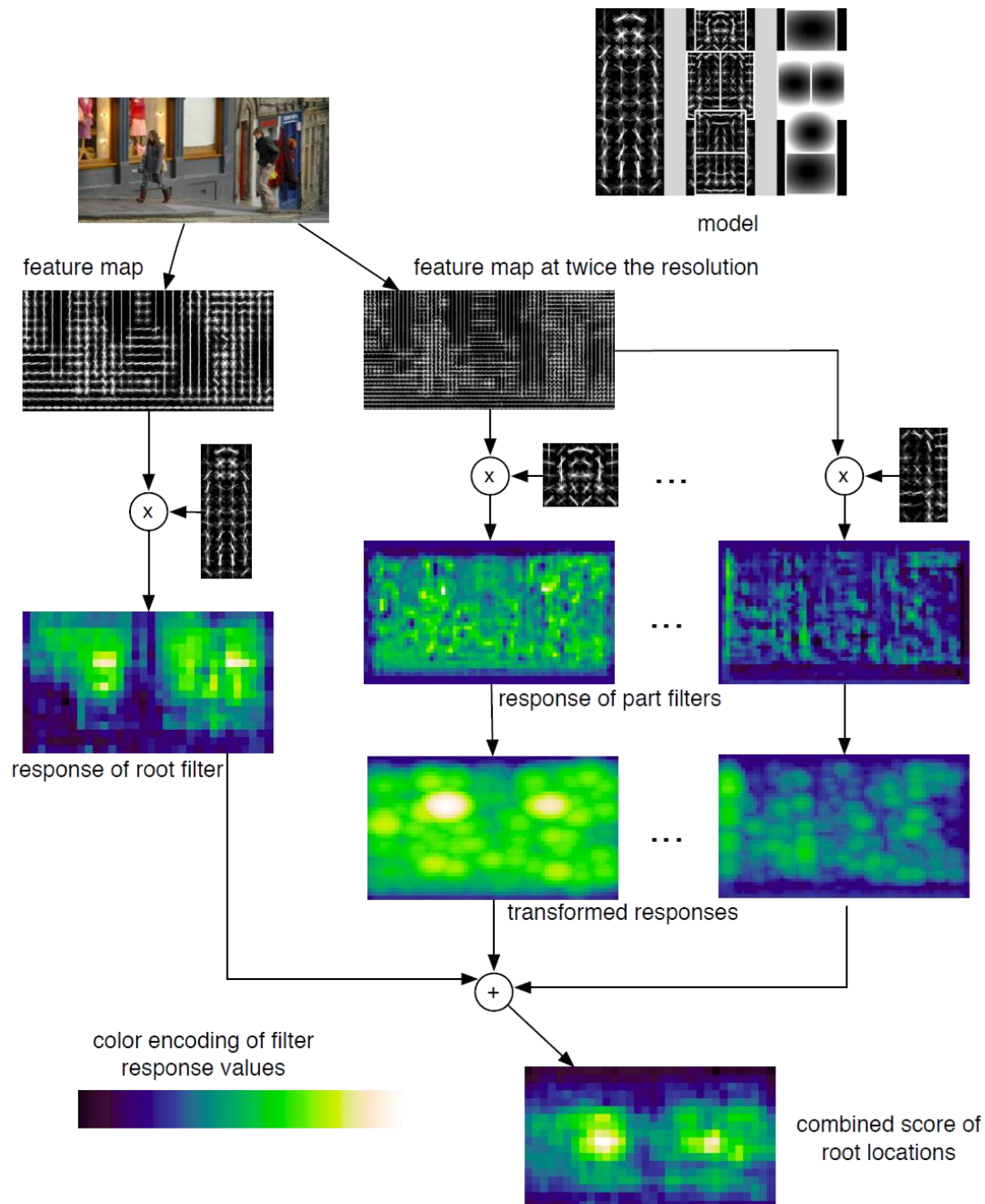
deformation parameters



$$\text{score}(z) = \beta \cdot \Psi(H, z)$$

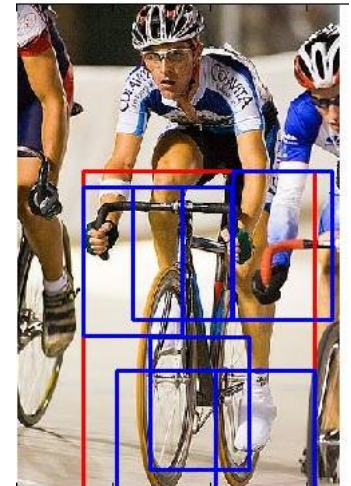
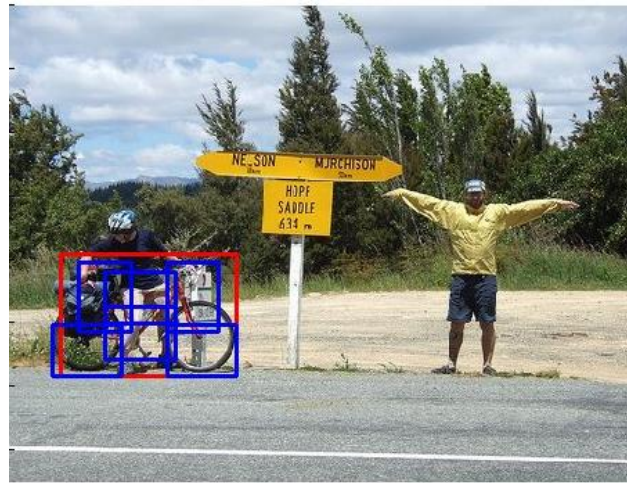
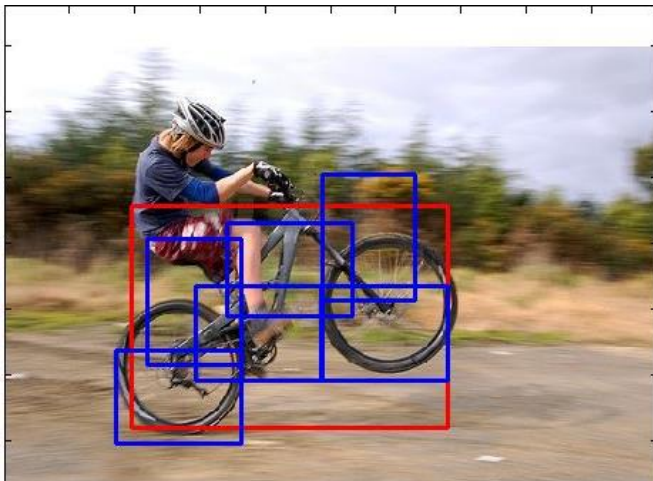
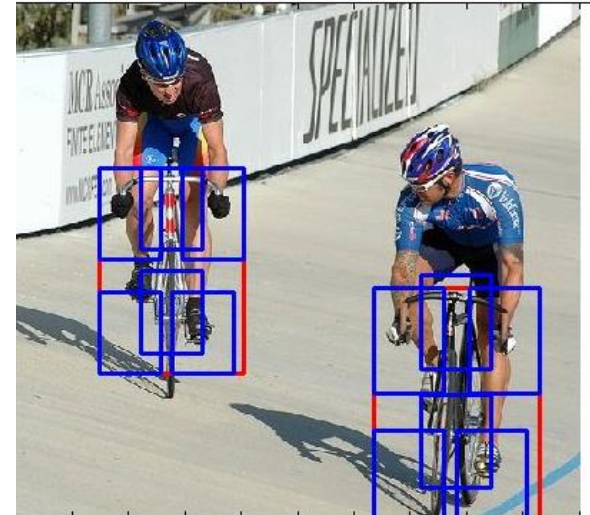
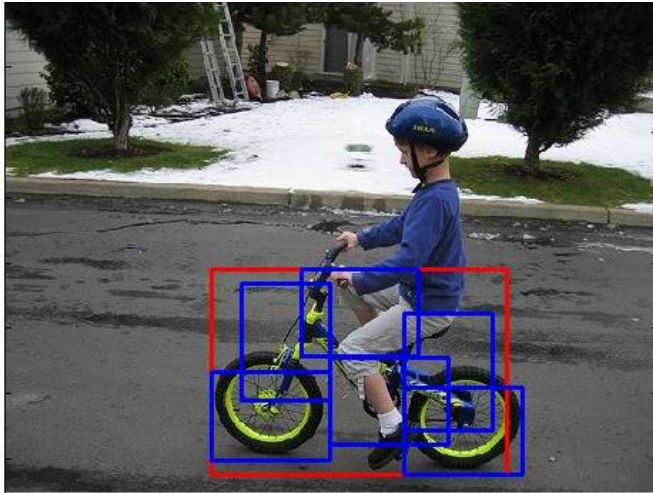
concatenation filters and
deformation parameters

concatenation of HOG
features and part
displacement features



Efficient computation
through Generalized
Distance Transform.

Results: Bicycles

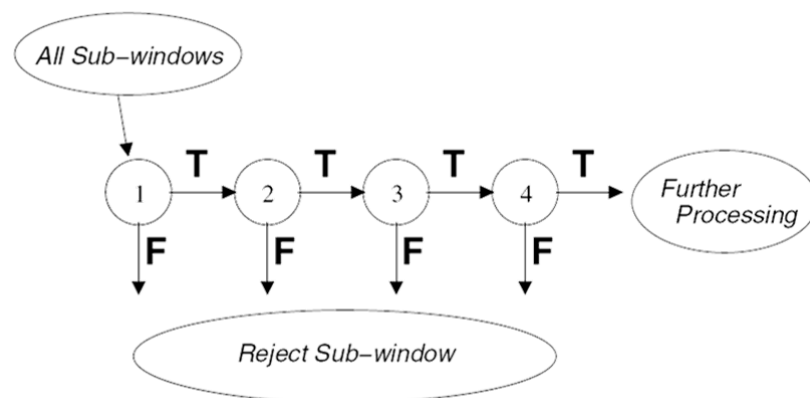


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Cascading Classifiers for Detection

- Even if the classifiers are fast, each new image has a lot of possible windows to search.
- For efficiency, apply less accurate but faster classifiers first to immediately discard windows that clearly appear to be negative; e.g.,
 - Filter for promising regions with an initial inexpensive classifier
 - Build a chain of classifiers, choosing cheap ones with low false negative rates early in the chain

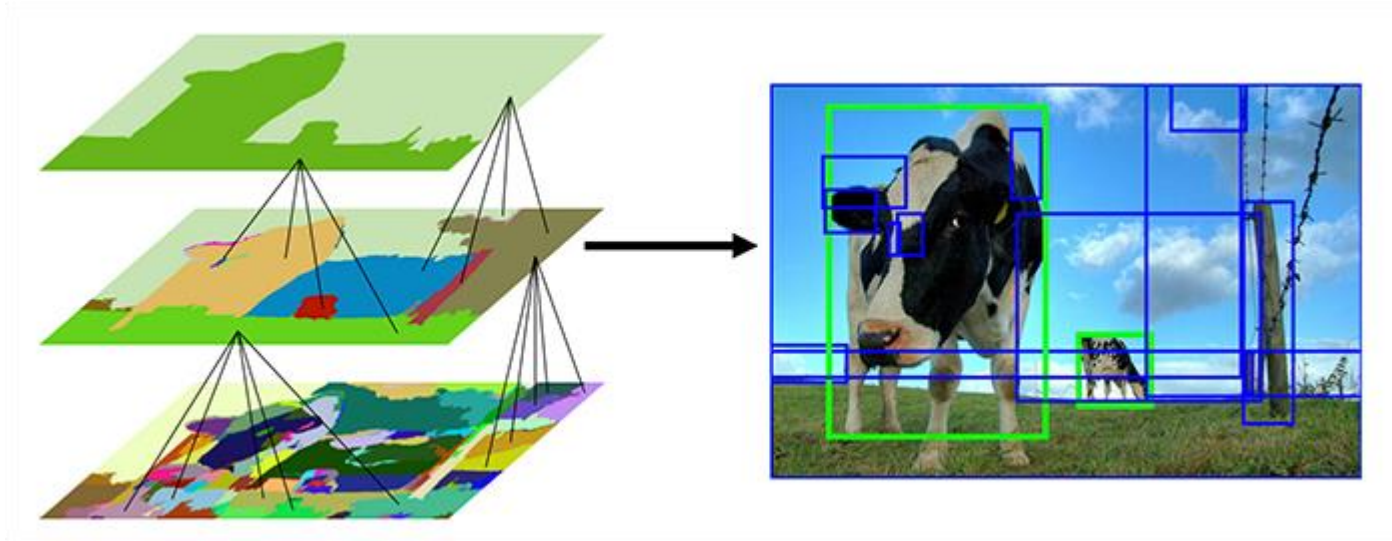


[Fleuret & Geman, IJCV 2001]

[Rowley et al., PAMI 1998]

[Viola & Jones, CVPR 2001]

Extension: Region Proposal Generation



- Alternative to exhaustive sliding-window search
 - E.g., **Selective Search** [Uijlings 2012]
 - Use bottom-up superpixel segmentation
 - Hierarchically group superpixels into larger proposal regions
 - Use those proposal regions to evaluate a classifier
 - Advantage: Focus effort on promising image regions
 - We can spend more computation on more powerful classifiers

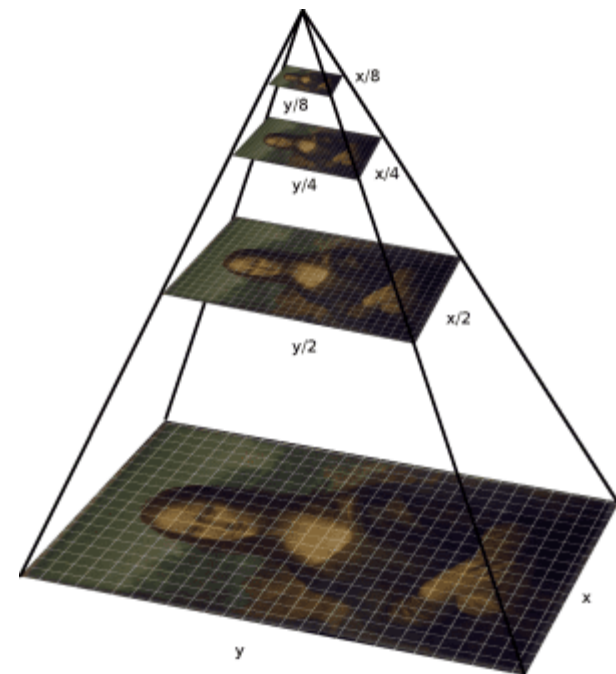
Summary: Sliding-Windows

- Pros

- Simple detection protocol to implement
- Good feature choices critical
- Past successes for certain classes
- Good detectors available

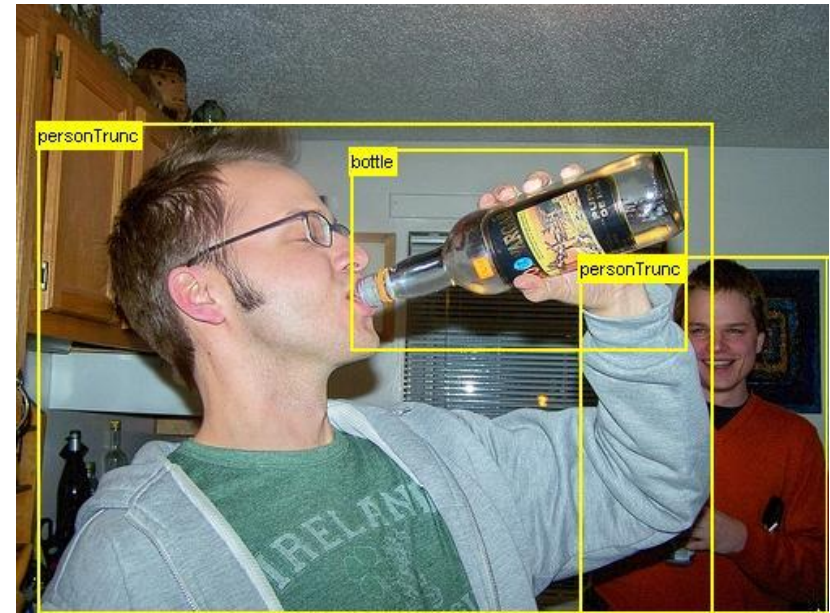
- Cons/Limitations

- High computational complexity
 - For example: 250,000 locations x 30 orientations x 4 scales = 30,000,000 evaluations!
 - This puts tight constraints on the classifiers we can use.
 - If training binary detectors independently, this means cost increases linearly with number of classes.
- With so many windows, false positive rate better be low



Limitations (continued)

- Not all objects are “box” shaped



Limitations (continued)

- If considering windows in isolation, context is lost



Sliding window



Detector's view

You Can Try It At Home...

- Deformable part-based models have been very successful in several evaluations.
 - ⇒ Approach was [state-of-the-art](#) until few years ago
- Code available
 - Both HOG and DPM are included in OpenCV
- References
 - N. Dalal, B. Triggs, [Histograms of Oriented Gradients for Human Detection](#), CVPR, 2005.
 - P.F. Felzenszwalb, R.B. Girshick, D. McAllester, D. Ramanan [Object Detection with Discriminatively Trained Part-Based Models](#), IEEE Trans. PAMI, Vol. 32(9), 2010.